CT-PO vs. SBAC Subtest Performance Indicators

Ji Zeng

Table of Contents

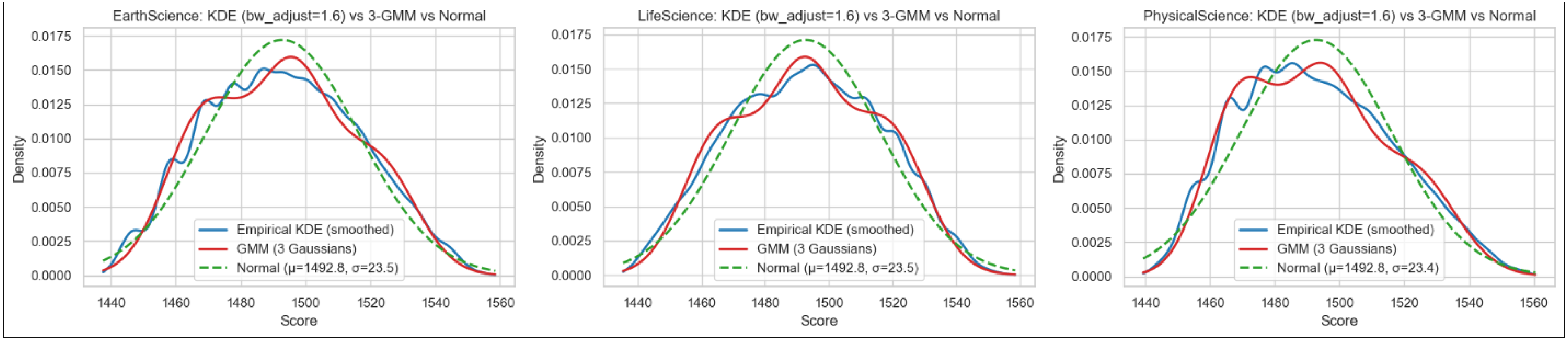
- ❑ Review of the empirical data explorations
- ❑ Report of the current pilot simulation study
  - o Simulation conditions
  - o Simulation results

Empirical Data Explorations

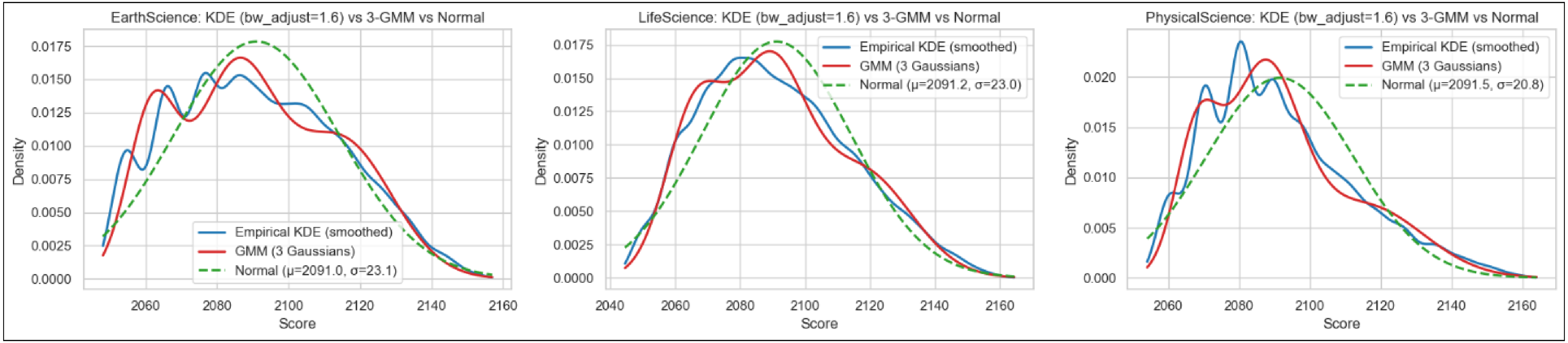
Values of Subscore Indicators

- Problems with the existing literature:
  - Uses of subscore reporting (not to predict the true performance on each subtest)
  - Multivariate normal distribution assumption (Sinharay, 2014; Gorney and Sinharay, 2025)

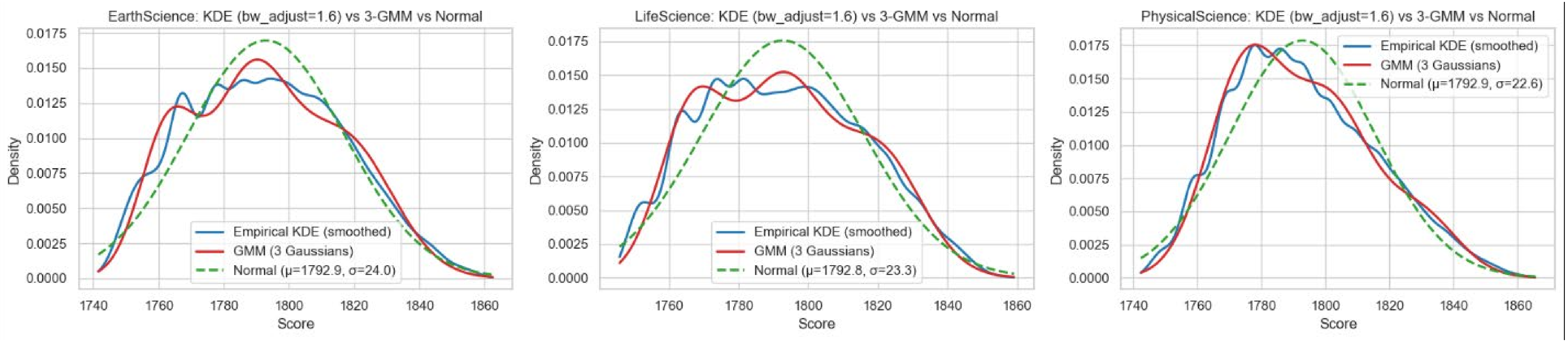
Real data distribution (based on scale scores)



Real data distribution (based on scale scores)



Real data distribution (based on scale scores)



CT-PO, SBAC, and K-modes

Grade	Subtest	K-modes		CT-PO		SBAC
		Cuts	Deviance	Cuts	Deviance	Deviance
5	ES	1481, 1507	71894.5	1485, 1507	71675.26	97032.84
	LS	1478, 1502		1481, 1507		
	PS	1488, 1512		1488, 1511		
8	ES	1785, 1812	73622.17	1783, 1811	71757.89	85159.56
	LS	1784, 1807		1784, 1808		
	PS	1785, 1810		1787, 1810		
11	ES	2087, 2110	61396.68	2088, 2105	57574.52	77622.43
	LS	2087, 2111		2091, 2113		
	PS	2089, 2109		2093, 2115		

CT-PO vs. SBAC

Performance Level	Grade 5		Grade 8		Grade 11	
	CT	SBAC	CT	SBAC	CT	SBAC
1	0	0	0	0	0	1329
2	4604	12184	5565	11488	1614	8793
3	2449	15101	4683	14475	2814	11706
4	0	1196	0	0	0	0

The current pilot simulation study

Simulation Procedures

- GMM→ true theta values (110 replications)
- Operational item parameters→0/1 responses
- Apply operational R2SS
- First 10 replications → Train CT-PO → Apply to the last 100 replications
- Computed classification accuracy (last 100 replications)
- Generated another set of 0/1 responses (100 replications)
- Computed classification consistency

Check Unidimensionality (Wainer et al., 2001)

- Compute Cronbach's coefficient α for each subtest
- Compute the variance-covariance matrix for each subtest using raw scores, denote it as $\mathbf{S}^{\text{obs}}$
- Compute the diagonal matrix as follows

$$\mathbf{D} = \begin{bmatrix} (1-\alpha_1)S_{11}^{\text{obs}} & 0 & 0 \\ 0 & (1-\alpha_2)S_{22}^{\text{obs}} & 0 \\ 0 & 0 & (1-\alpha_3)S_{33}^{\text{obs}} \end{bmatrix}$$

- Obtain $\mathbf{S}^{\text{true}}$ by computing $\mathbf{S}^{\text{true}} = \mathbf{S}^{\text{obs}} - \mathbf{D}$
- Then conduct eigenvalue decomposition of $\mathbf{S}^{\text{true}}$ to verify if a dominating single eigenvalue exists. If so, then the overall test can be modeled as unidimensional.

Batch 1 Simulation

Simulation Conditions: Batch 1

- Each subtest theta is simulated from a GMM with 3 normal distributions based on the following conditions:
  - Weights of data from each normal
  - Mean vectors
  - SD vectors
  - Correlations among three subtests

Simulation Conditions: Batch 1

- Weights of data from each normal:
  - [0.5, 0.3, 0.2]
  - [0.4, 0.4, 0.2]
  - [0.3, 0.4, 0.3]
  - [0.2, 0.4, 0.4]
  - [0.2, 0.5, 0.3]
  - [0.4, 0.3, 0.3]

Simulation Conditions: Batch 1

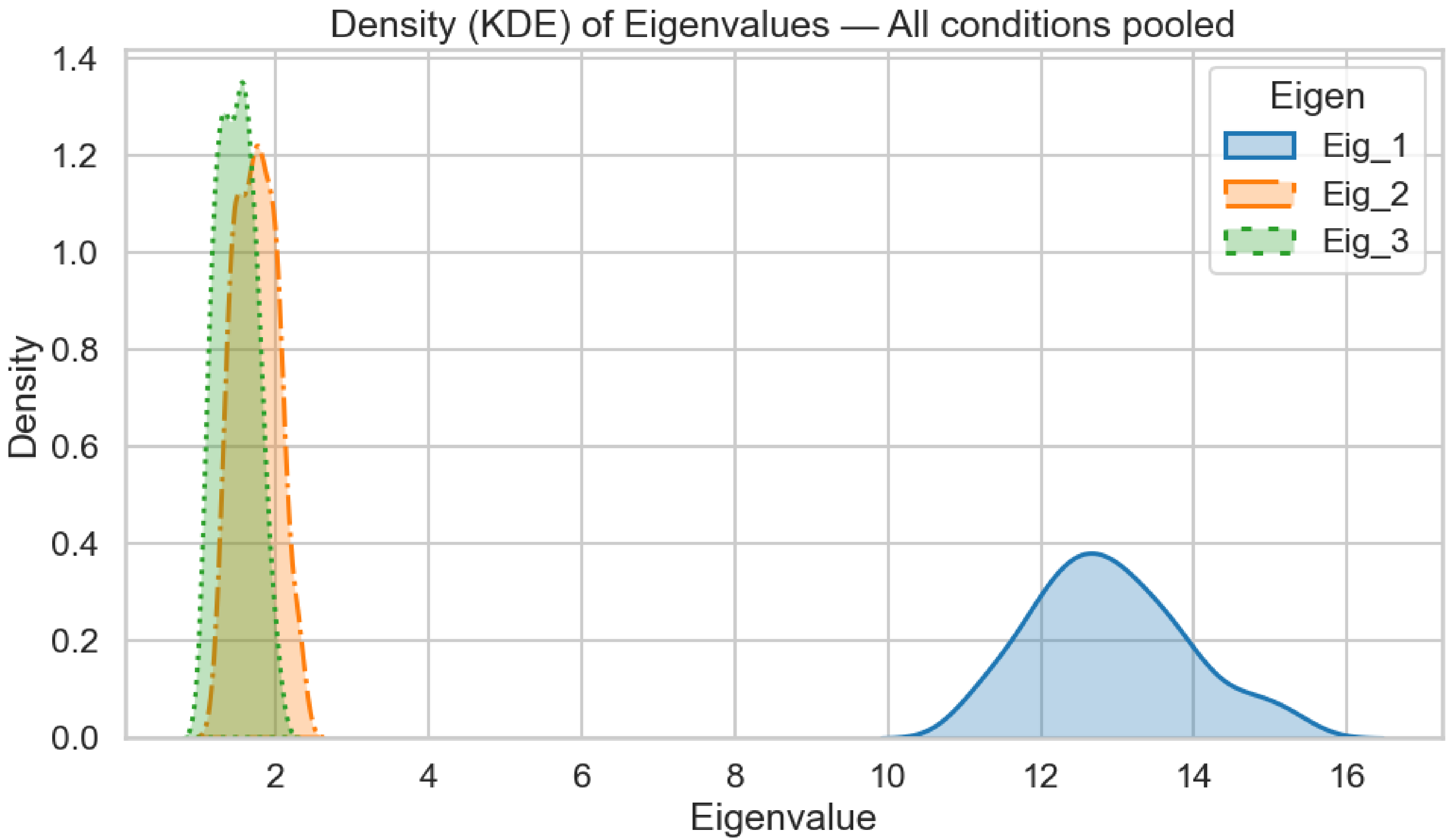
- Mean vectors:
  - [-0.85, 0.15, 1.15]
  - [-1.05, -0.15, 0.95]
  - [-0.95, 0.05, 1.05]

Simulation Conditions: Batch 1

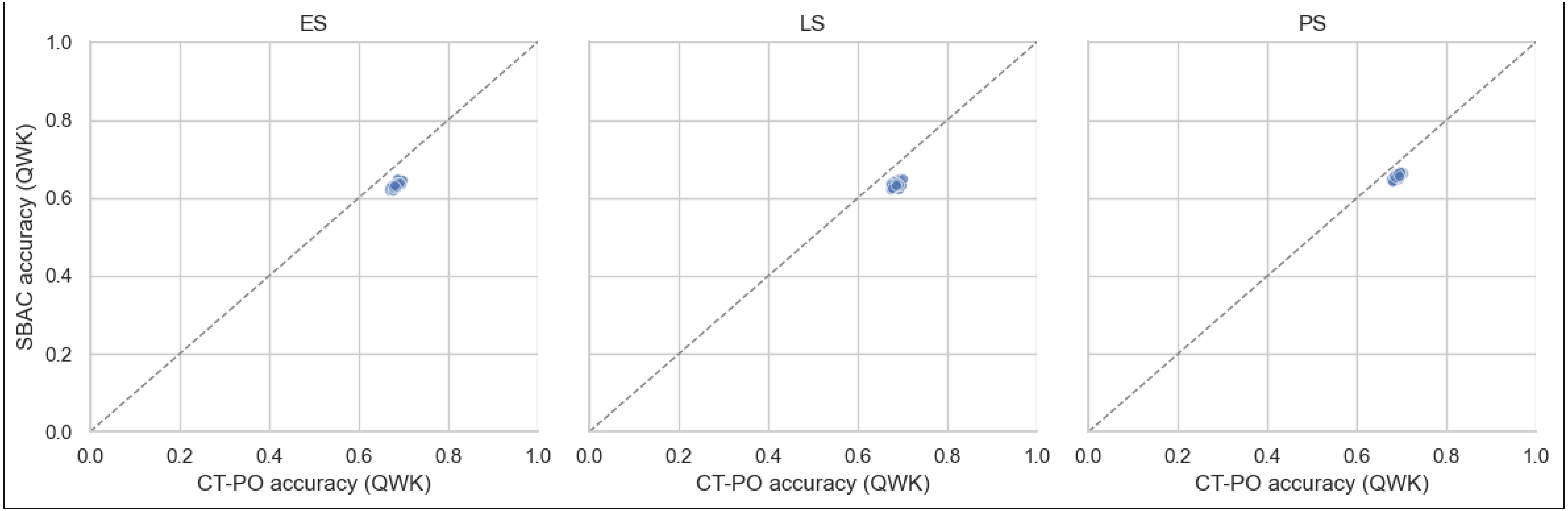
- SD vectors:
  - [0.45, 0.45, 0.45]
  - [0.40, 0.45, 0.50]

- Correlation among three subtests:
  - 0.65
  - 0.70
  - 0.75

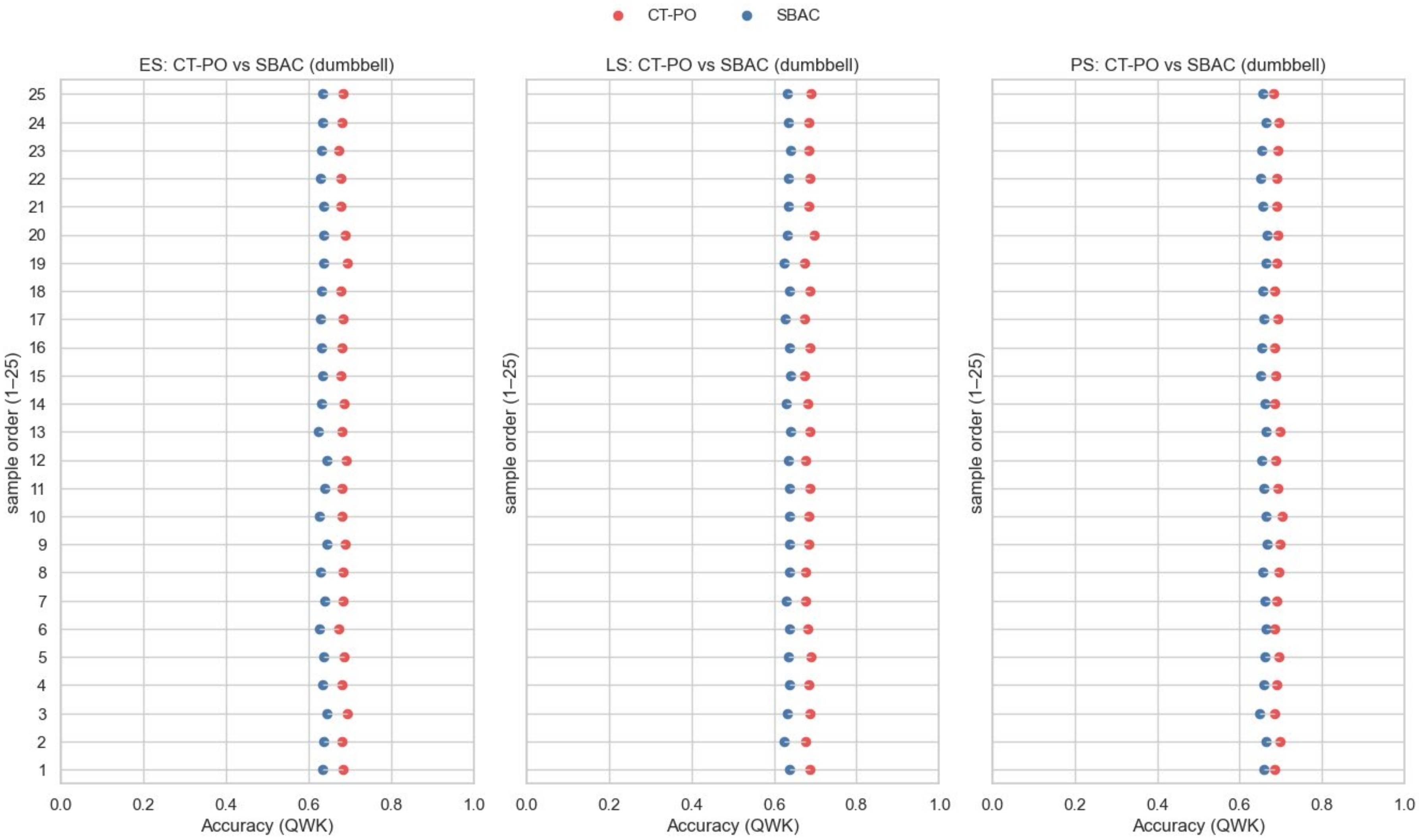
Checking simulation data unidimensionality



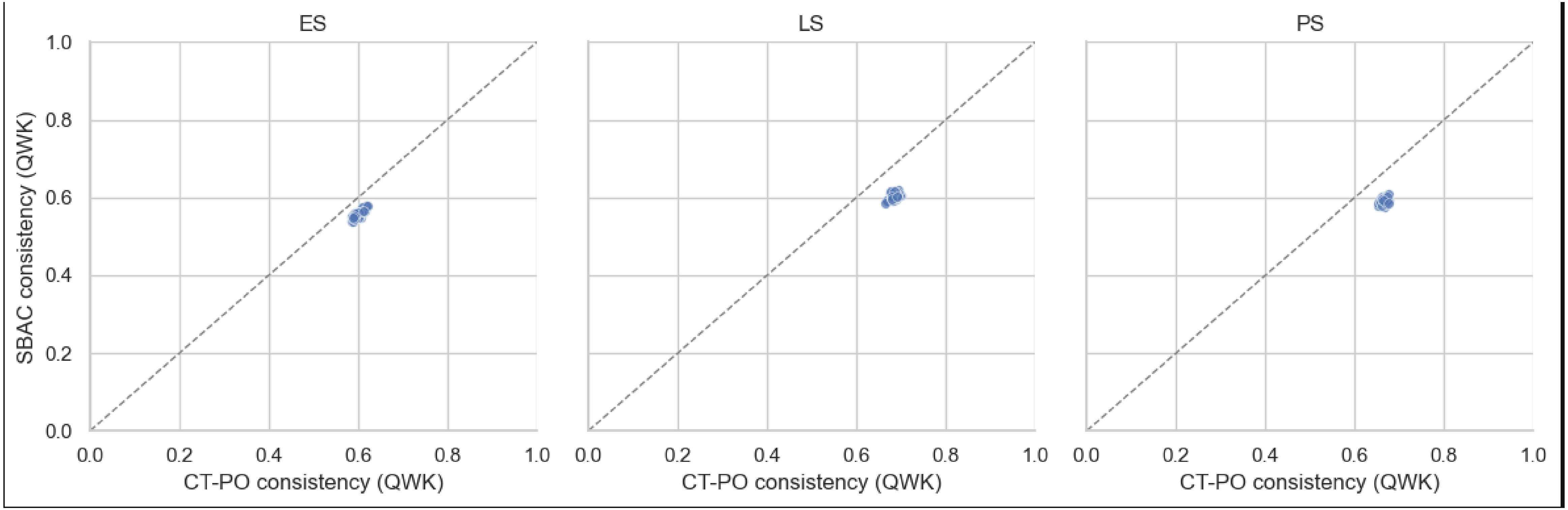
Within Condition Results: Accuracy QWK



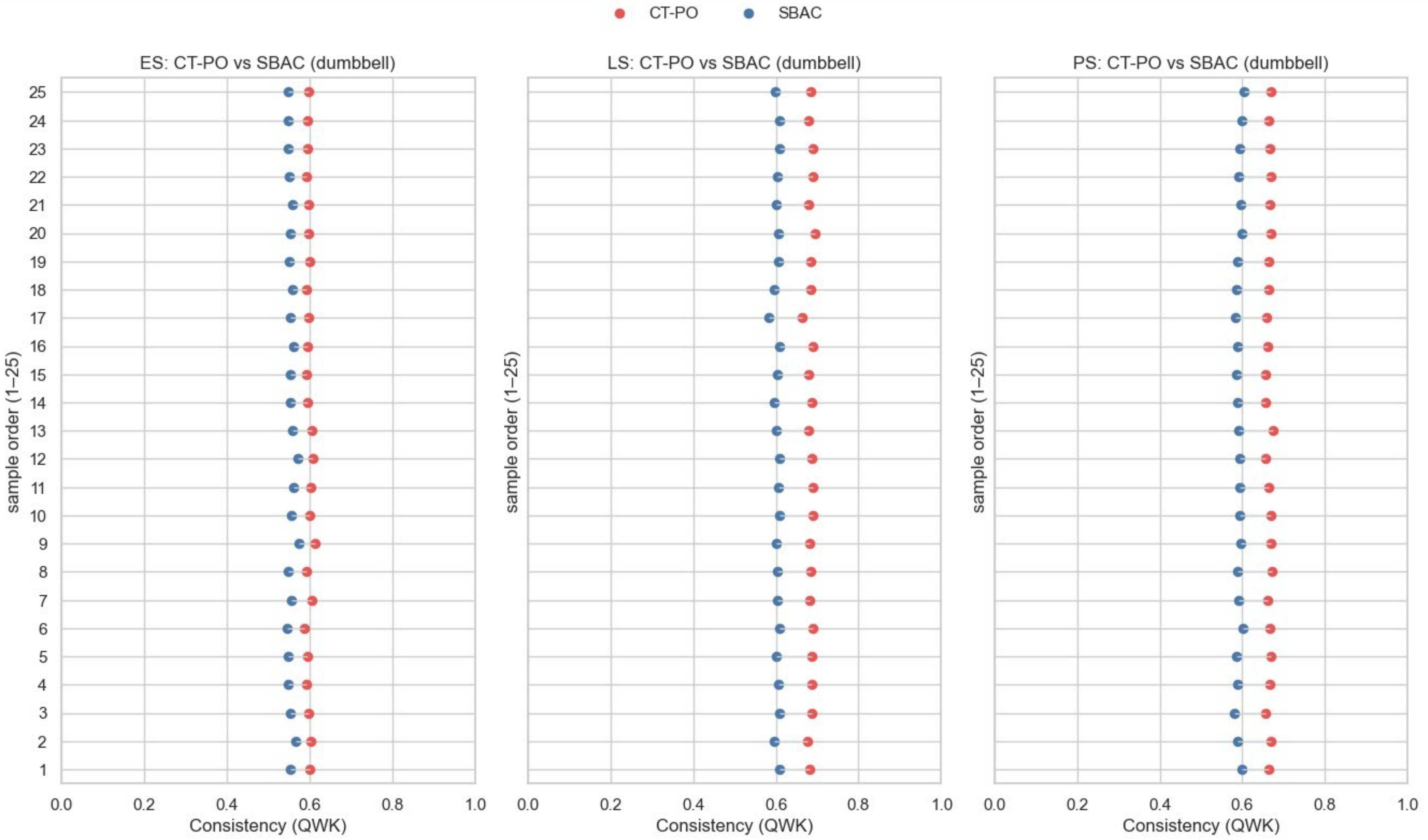
Within Condition Results: Accuracy QWK



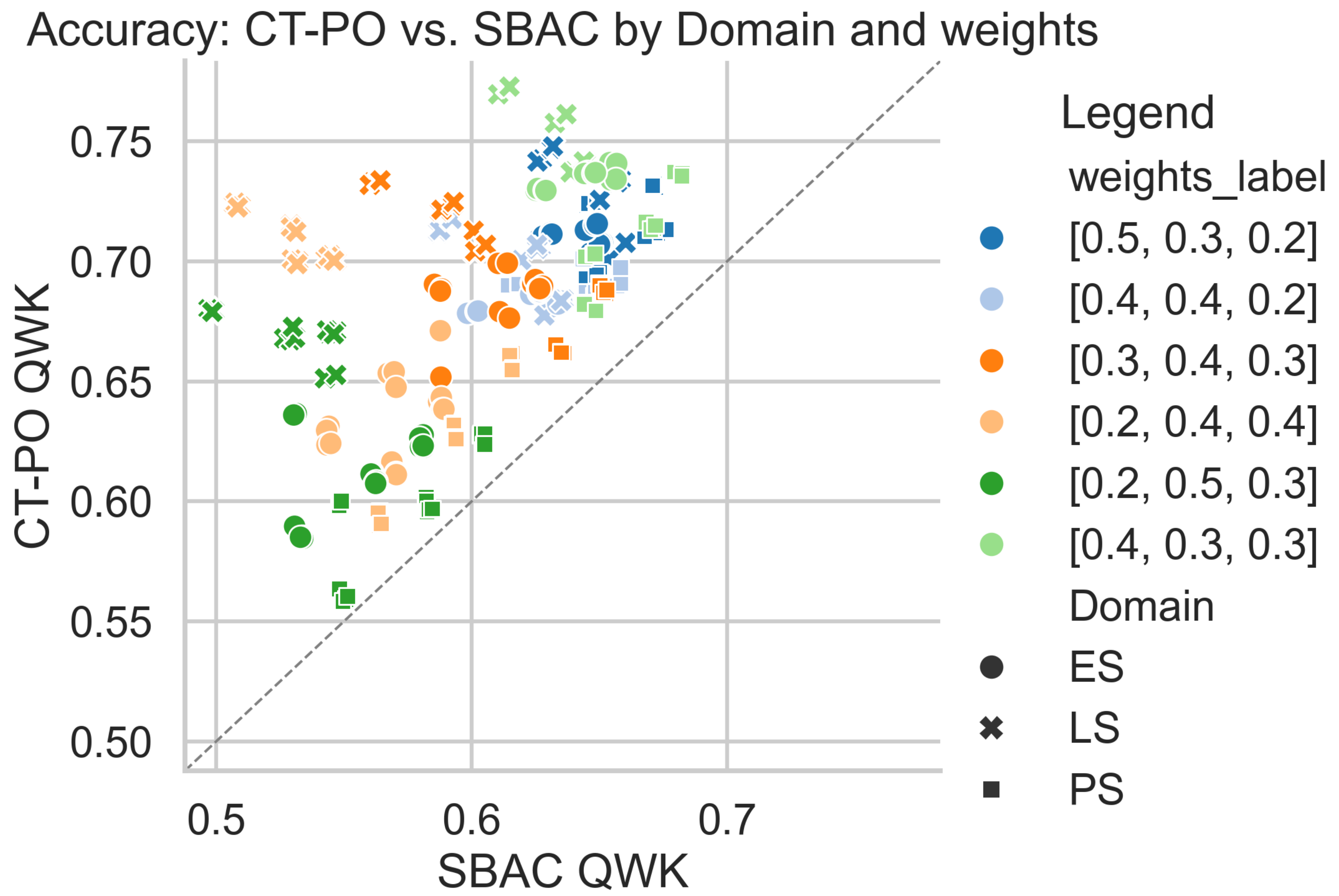
Within Condition Results: Consistency QWK



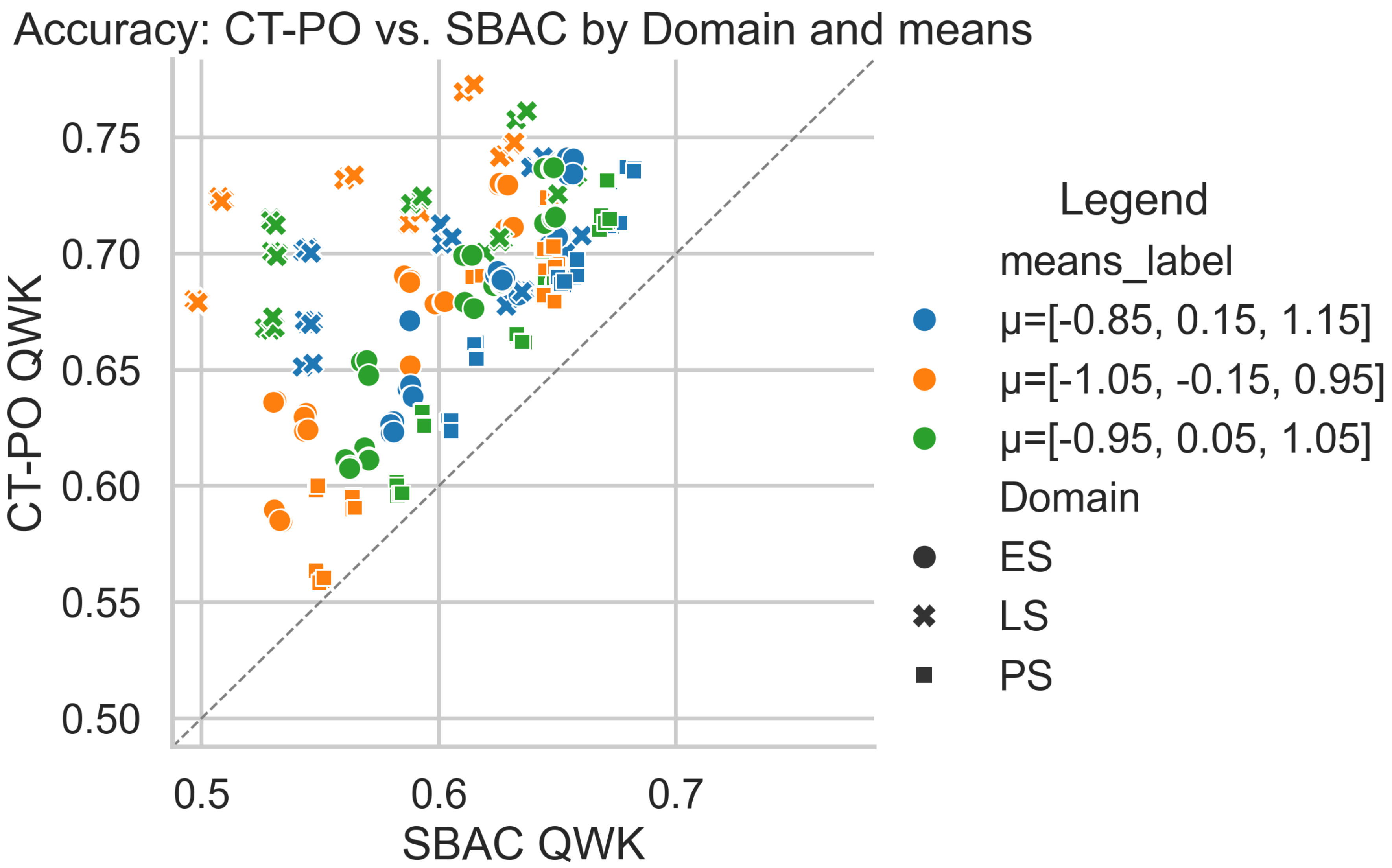
Within Condition Results: Consistency QWK



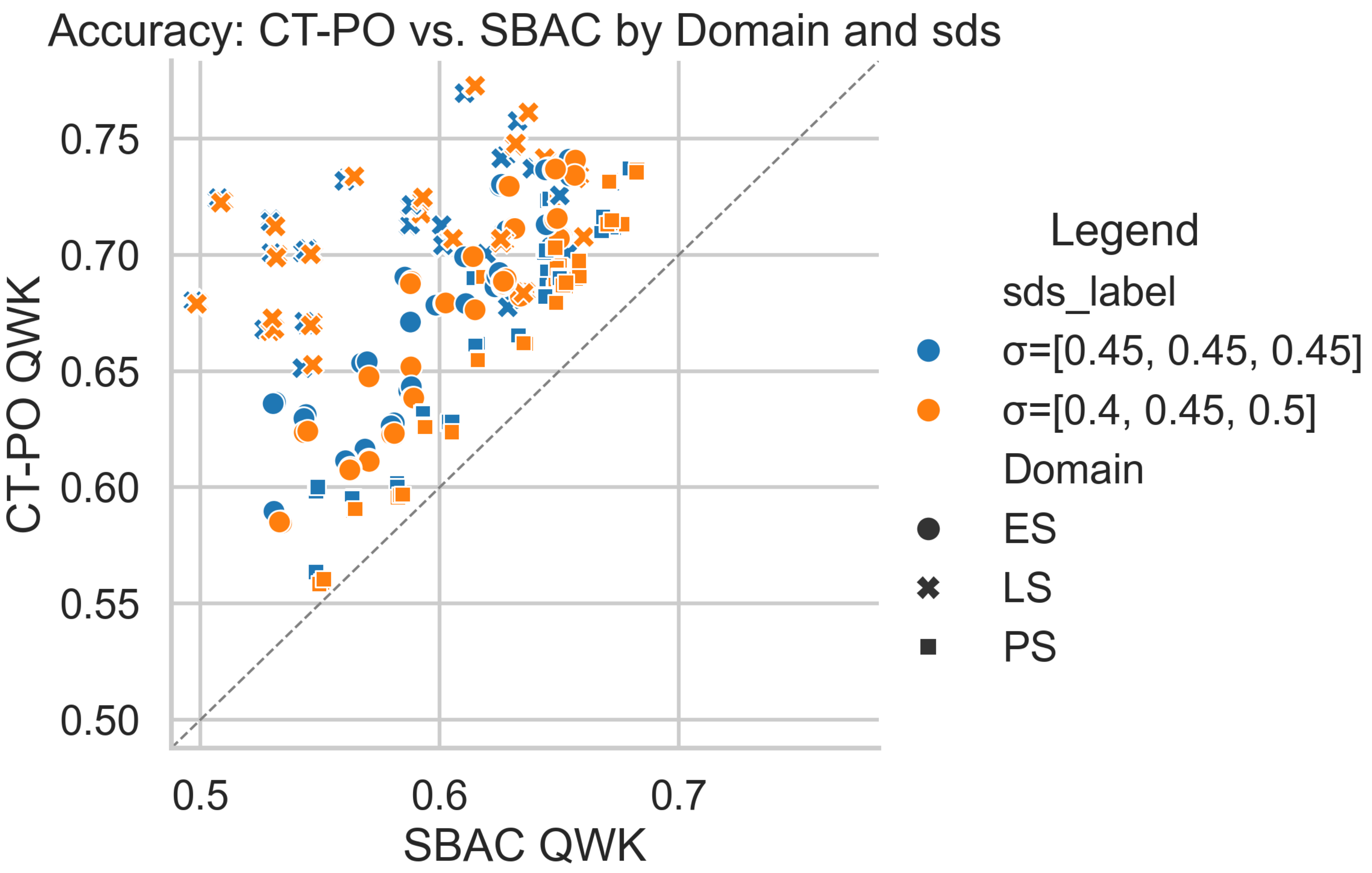
Accuracy: CT-PO vs. SBAC



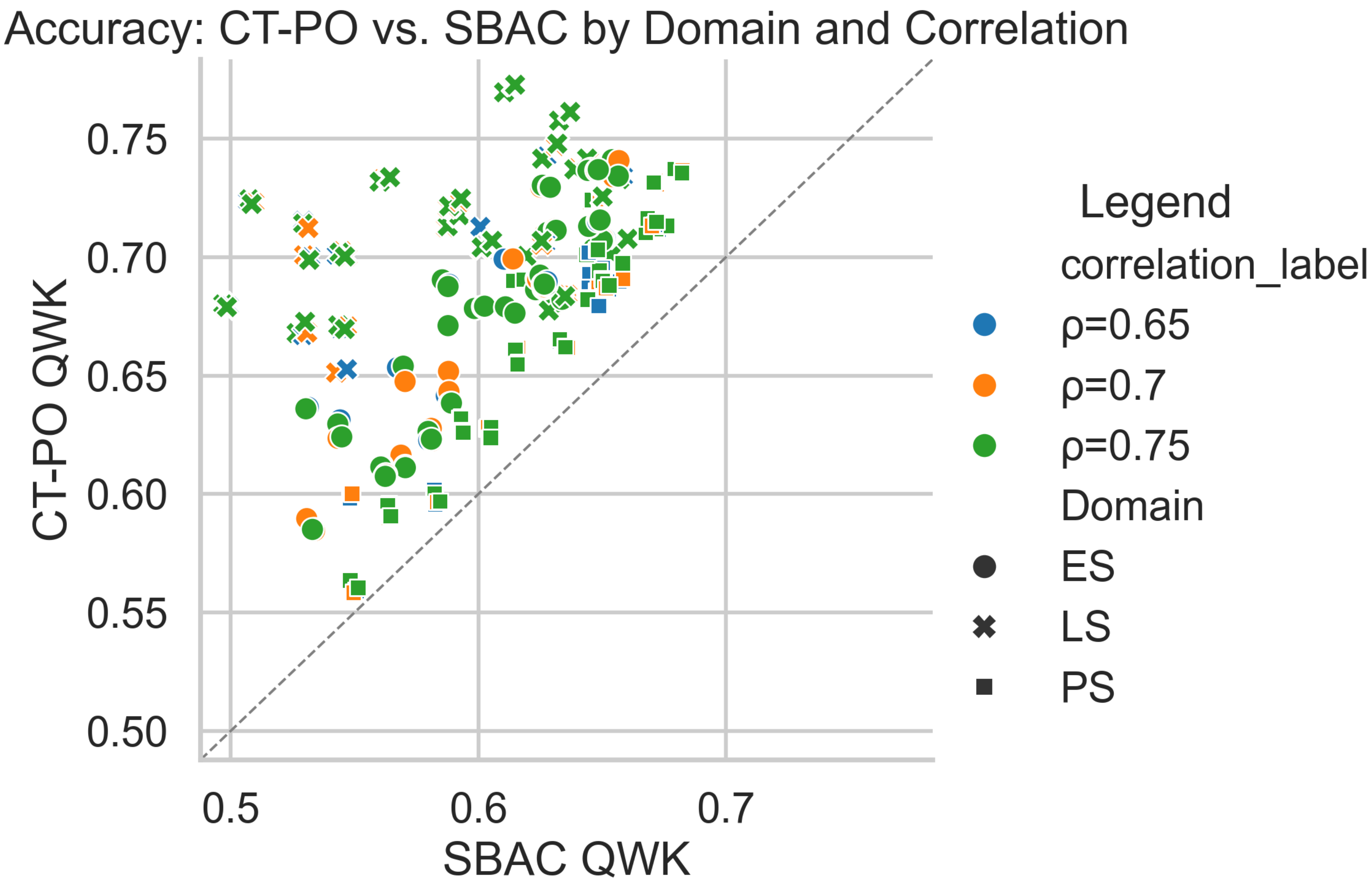
Accuracy: CT-PO vs. SBAC



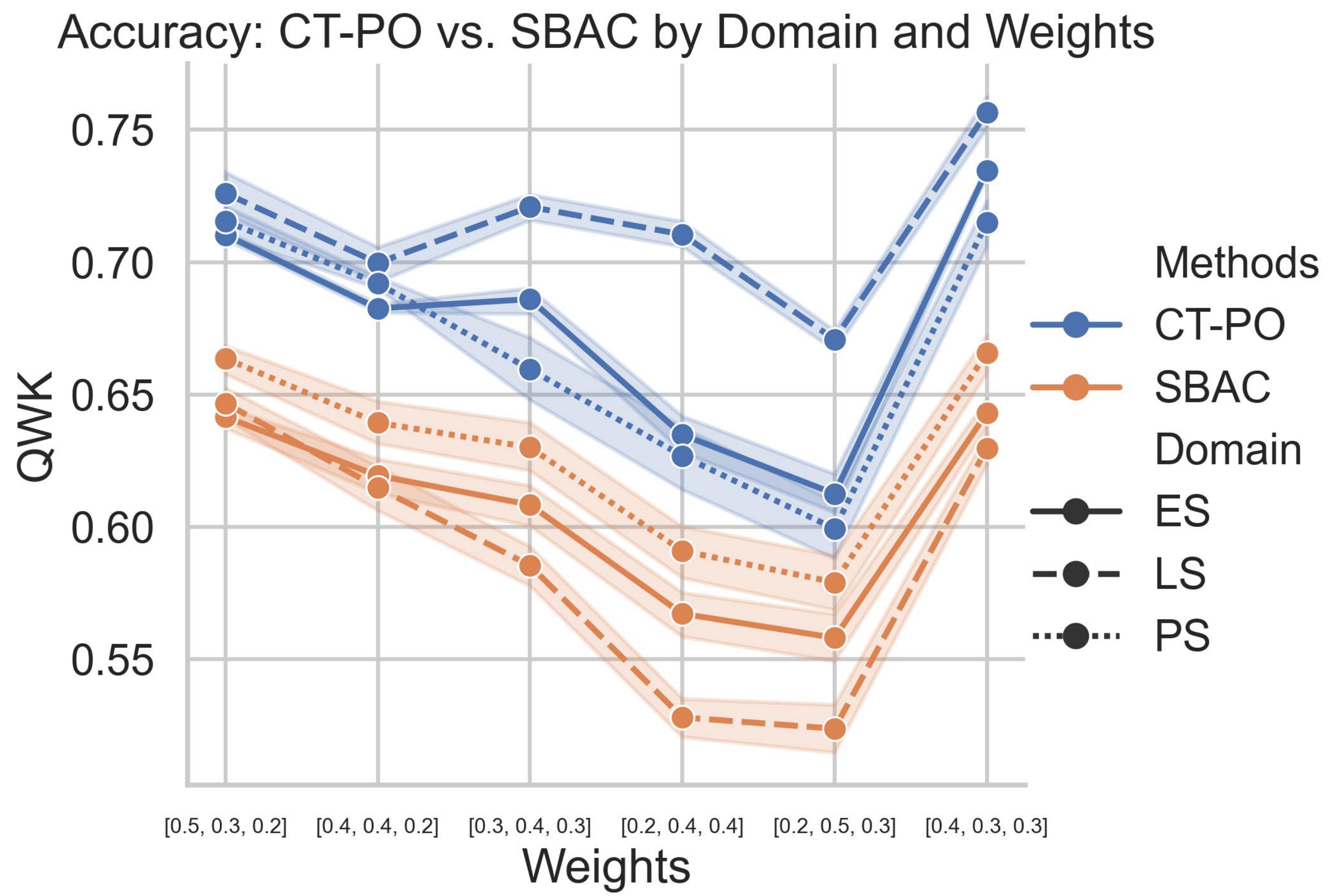
Accuracy: CT-PO vs. SBAC



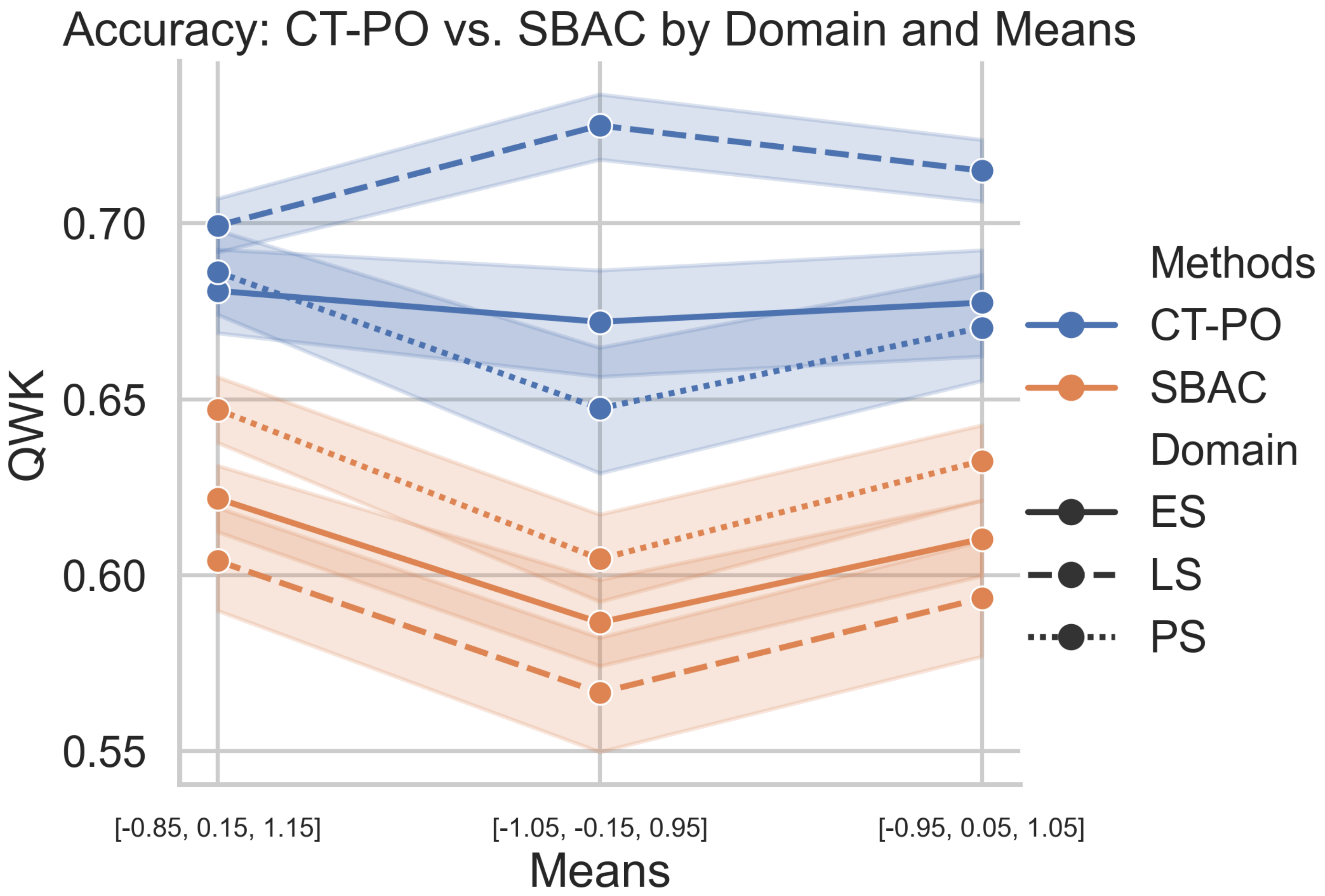
Accuracy: CT-PO vs. SBAC



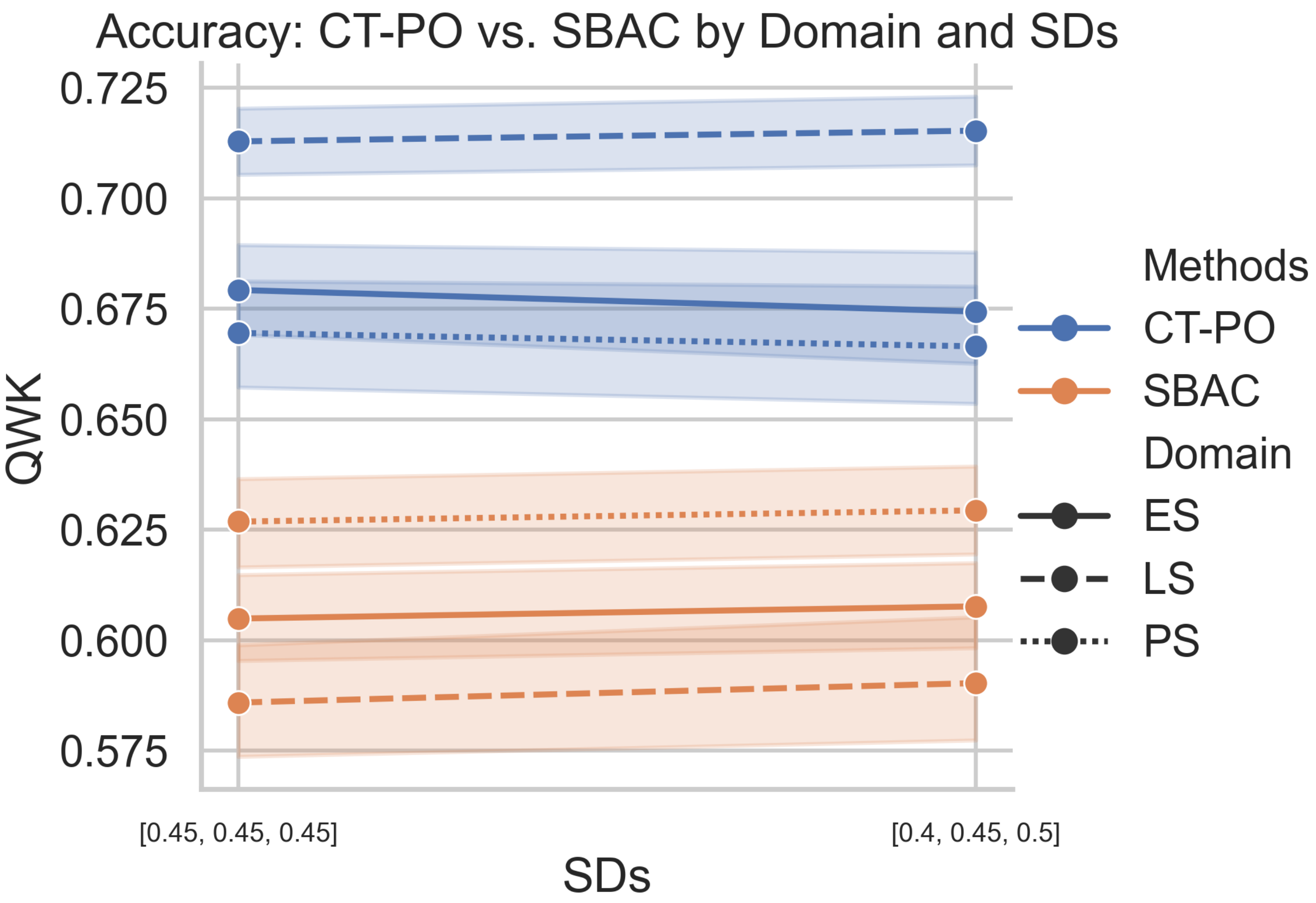
Accuracy: CT-PO vs. SBAC



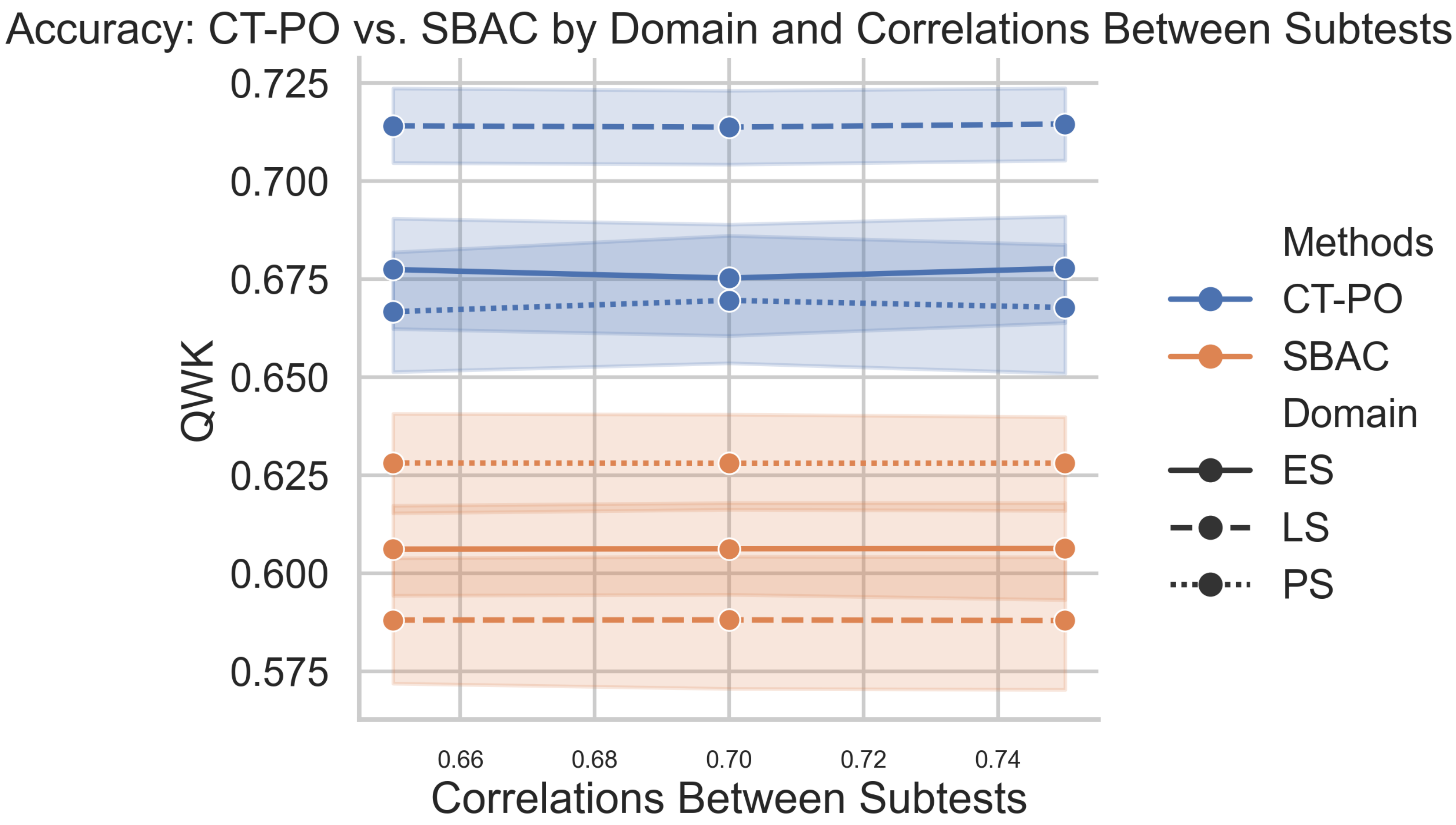
Accuracy: CT-PO vs. SBAC



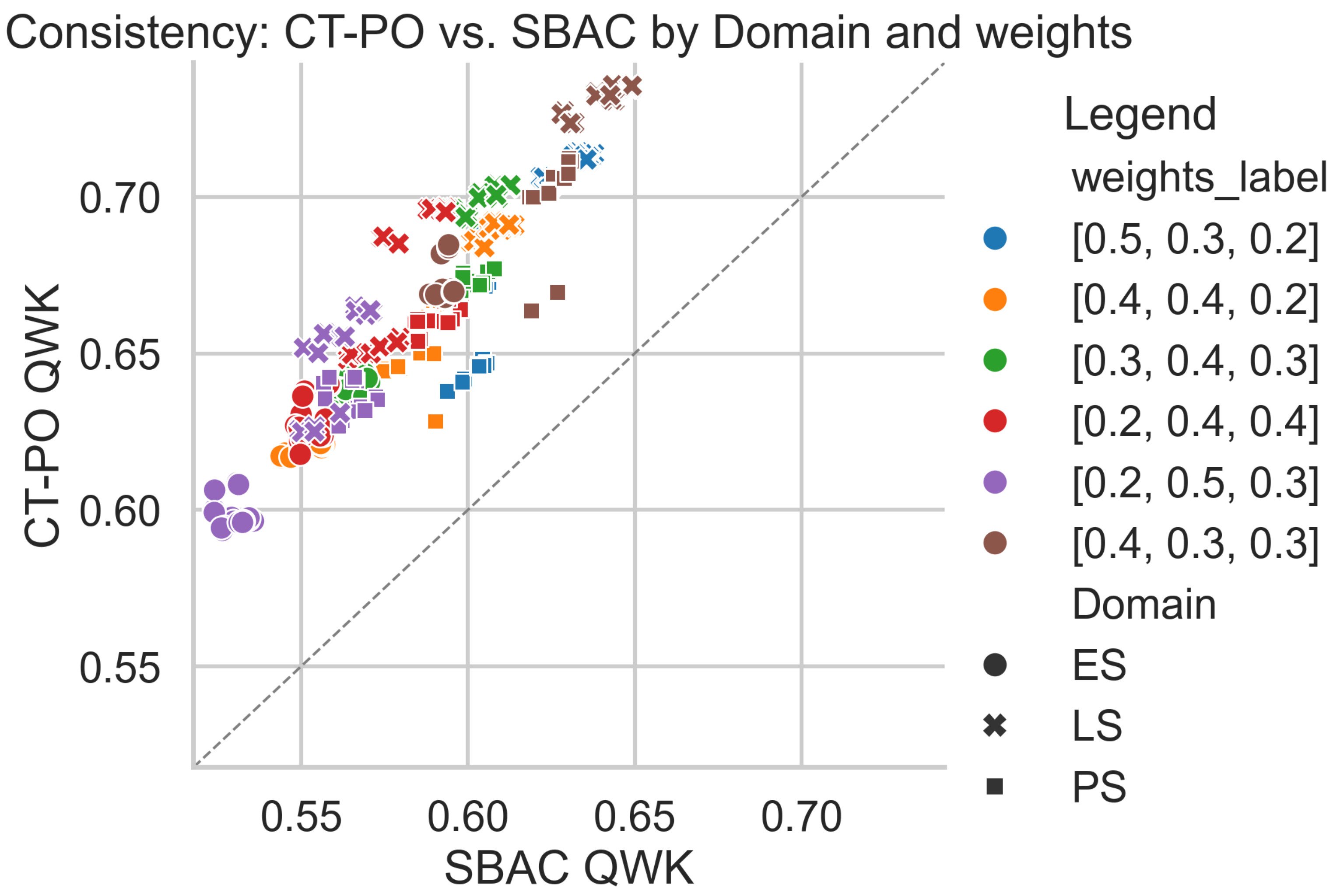
Accuracy: CT-PO vs. SBAC



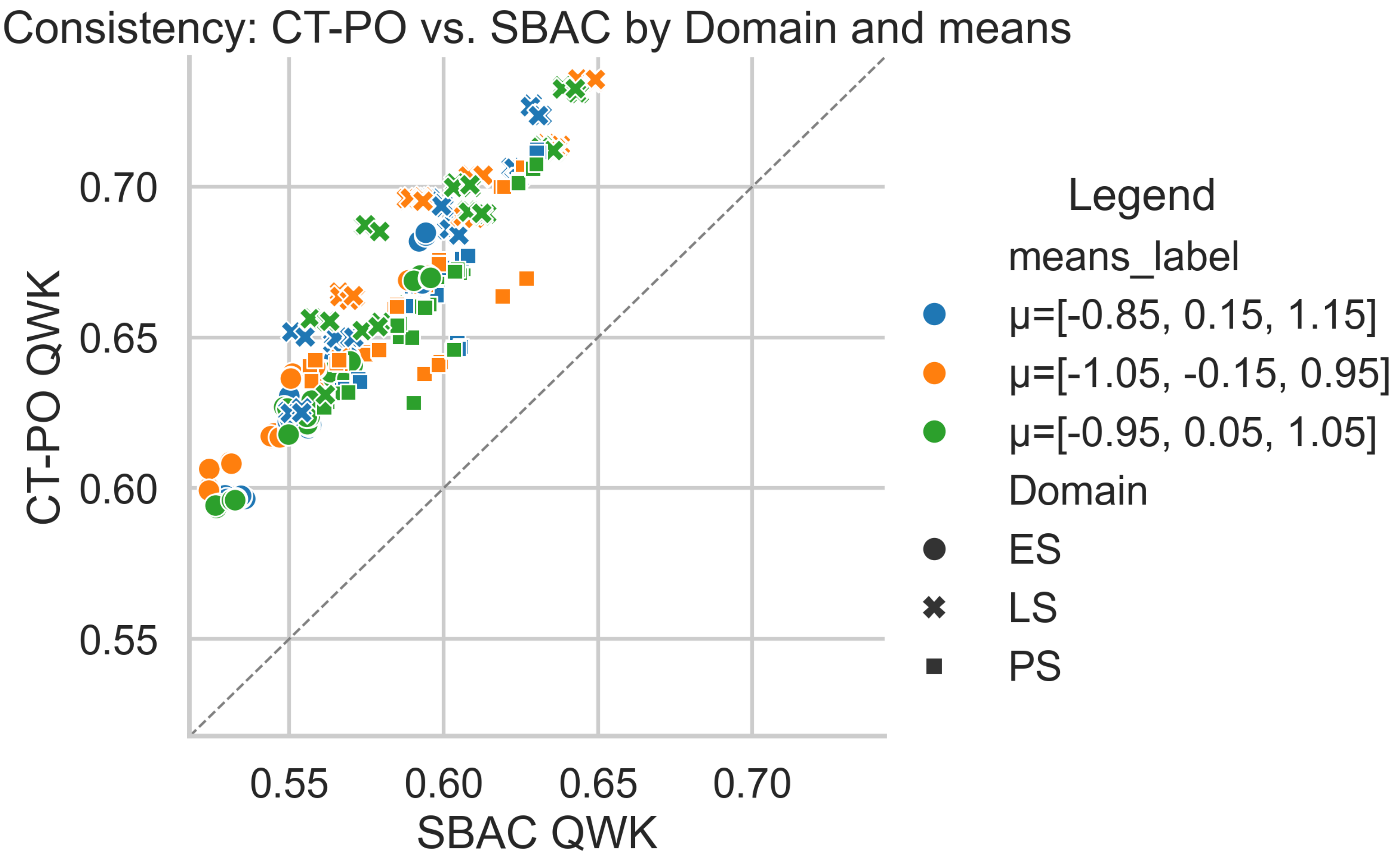
Accuracy: CT-PO vs. SBAC



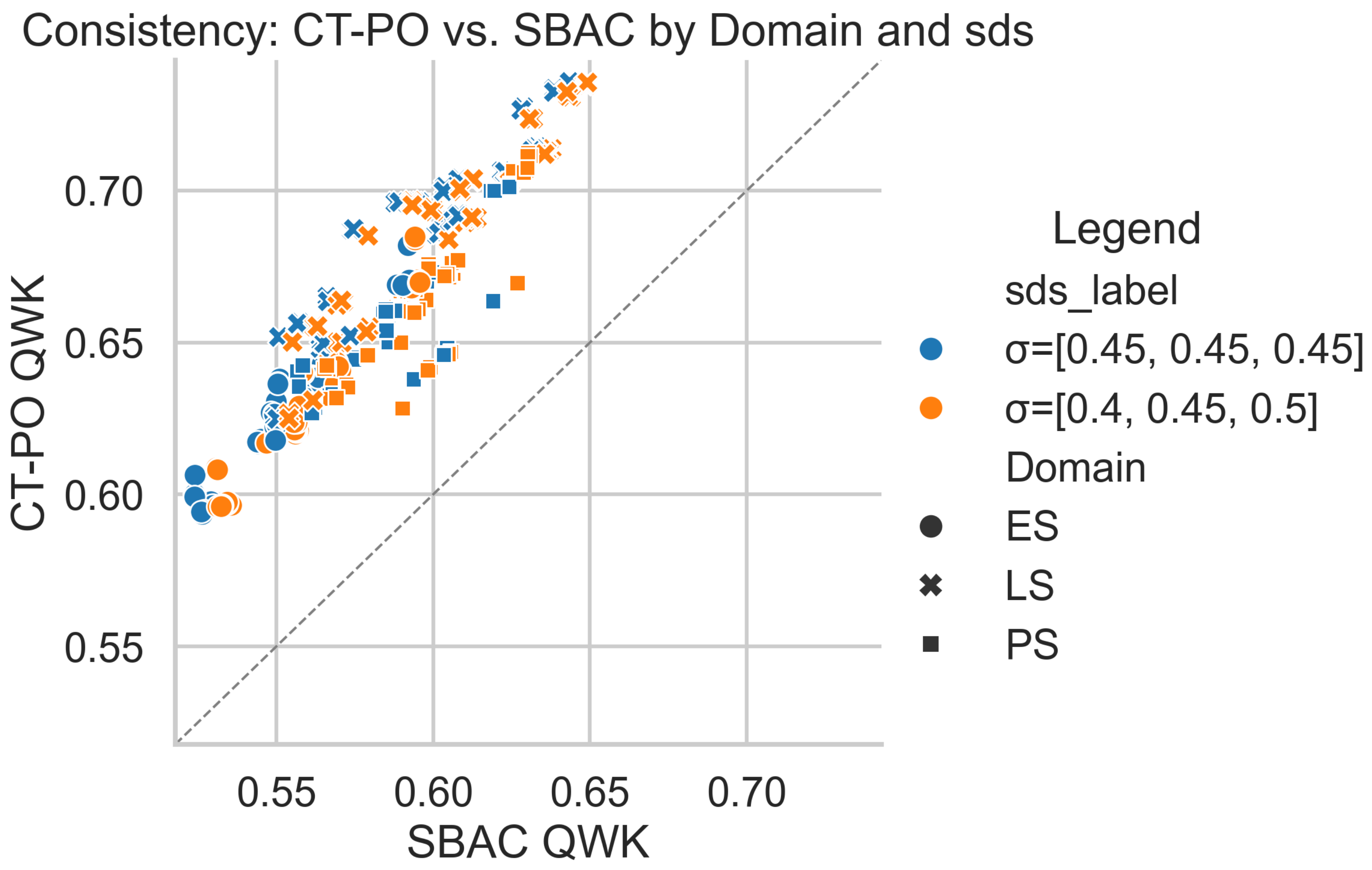
Consistency: CT-PO vs. SBAC



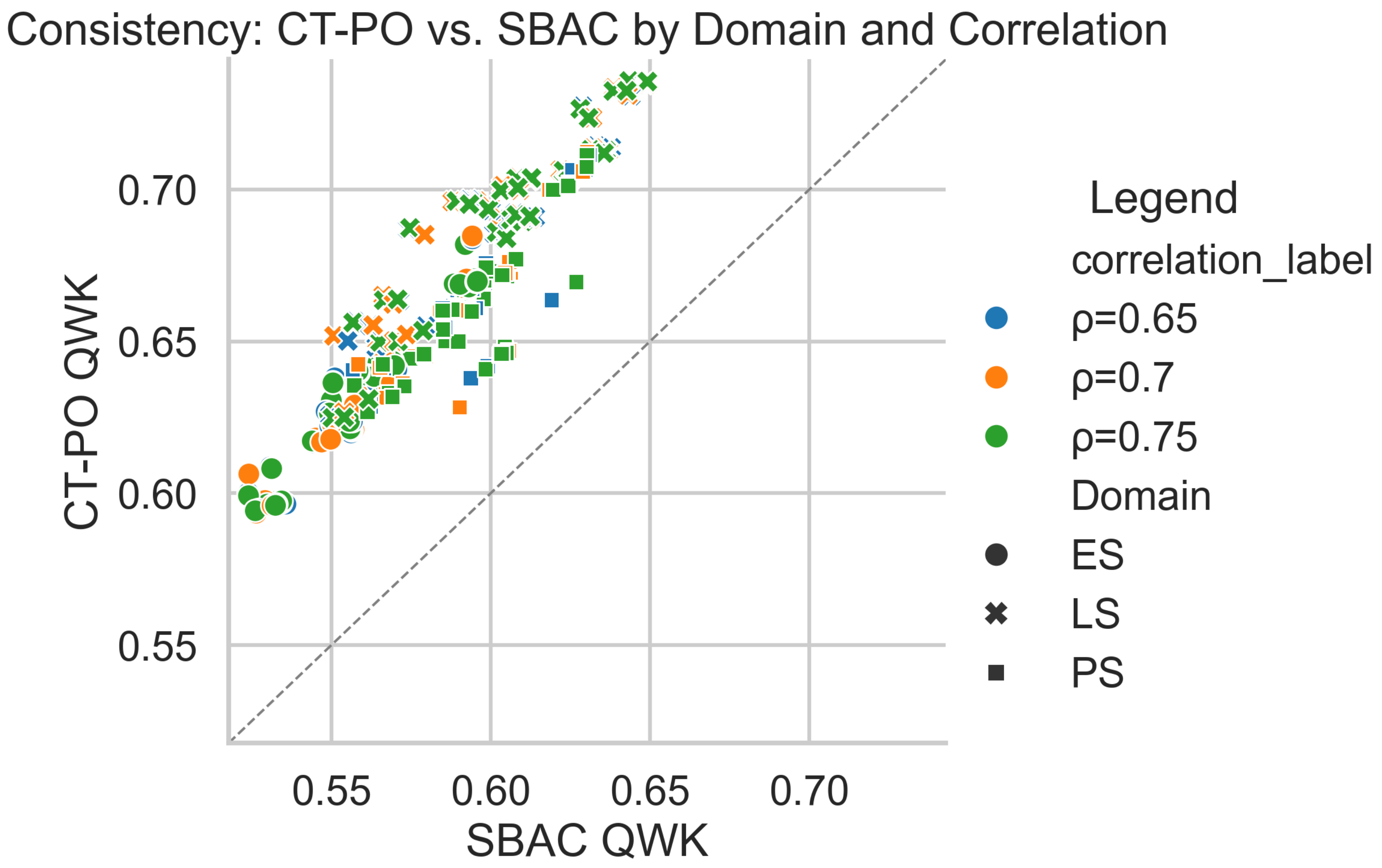
Consistency: CT-PO vs. SBAC



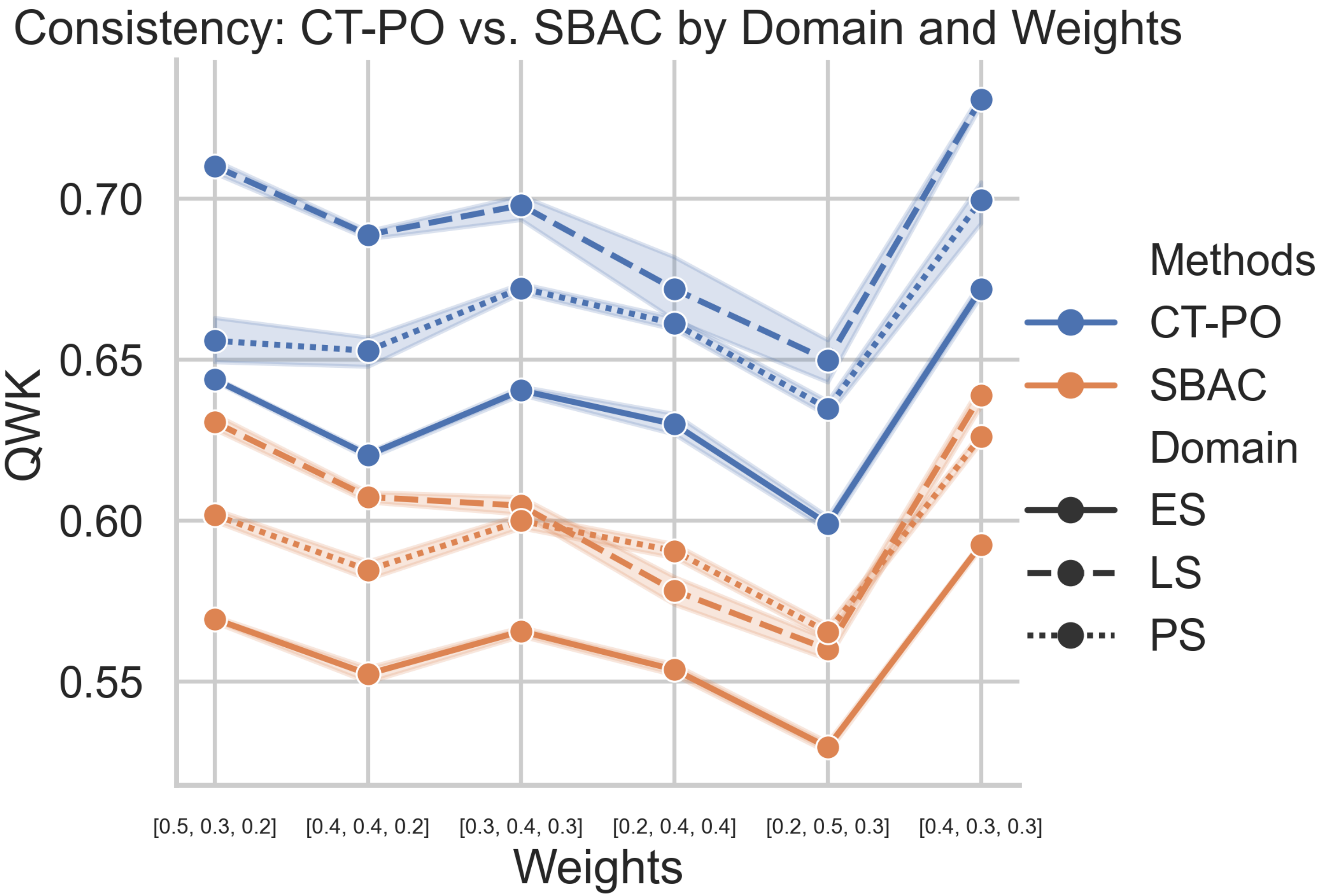
Consistency: CT-PO vs. SBAC



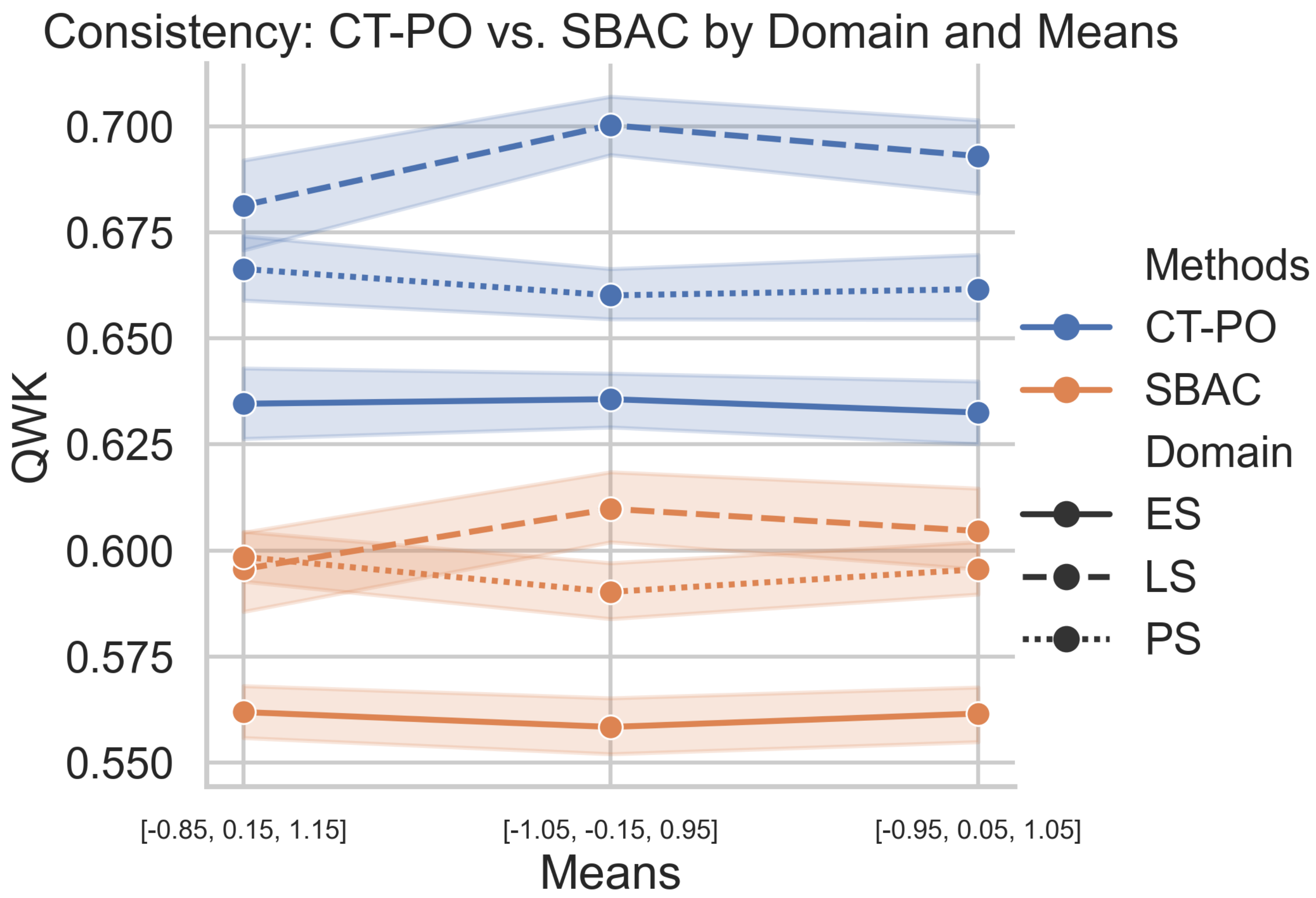
Consistency: CT-PO vs. SBAC



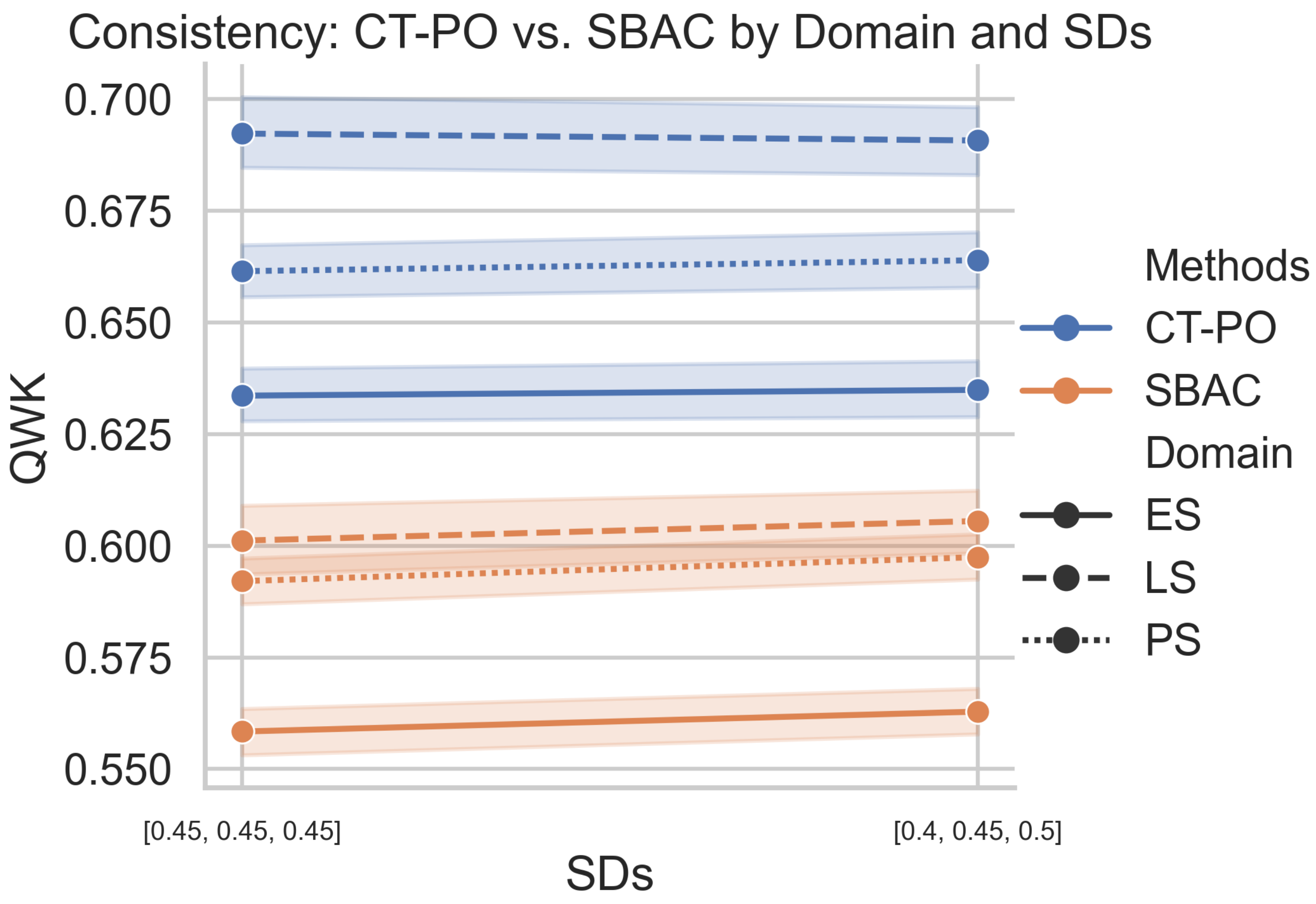
Consistency: CT-PO vs. SBAC



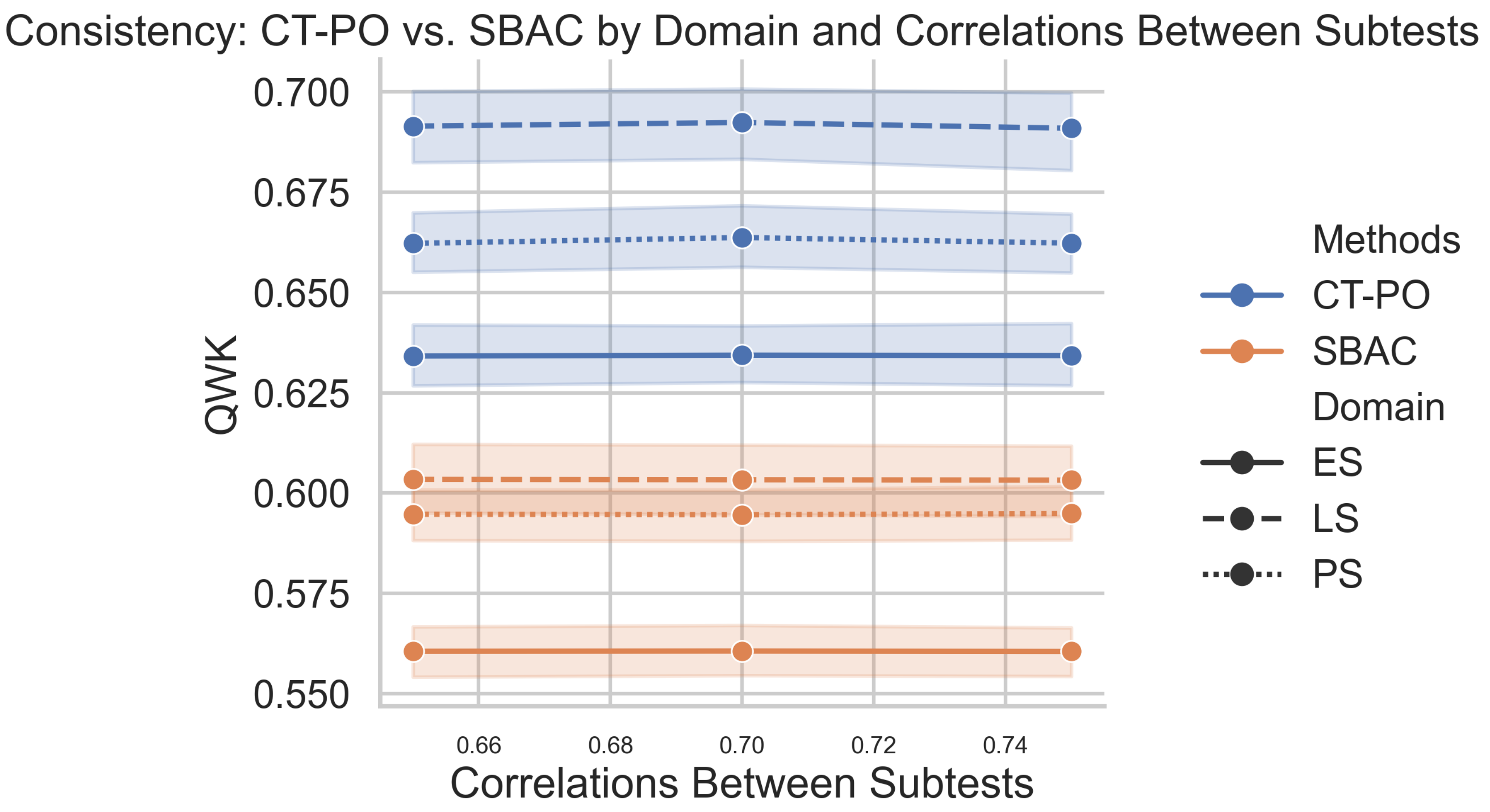
Consistency: CT-PO vs. SBAC



Consistency: CT-PO vs. SBAC



Consistency: CT-PO vs. SBAC



Batch 2 Simulation

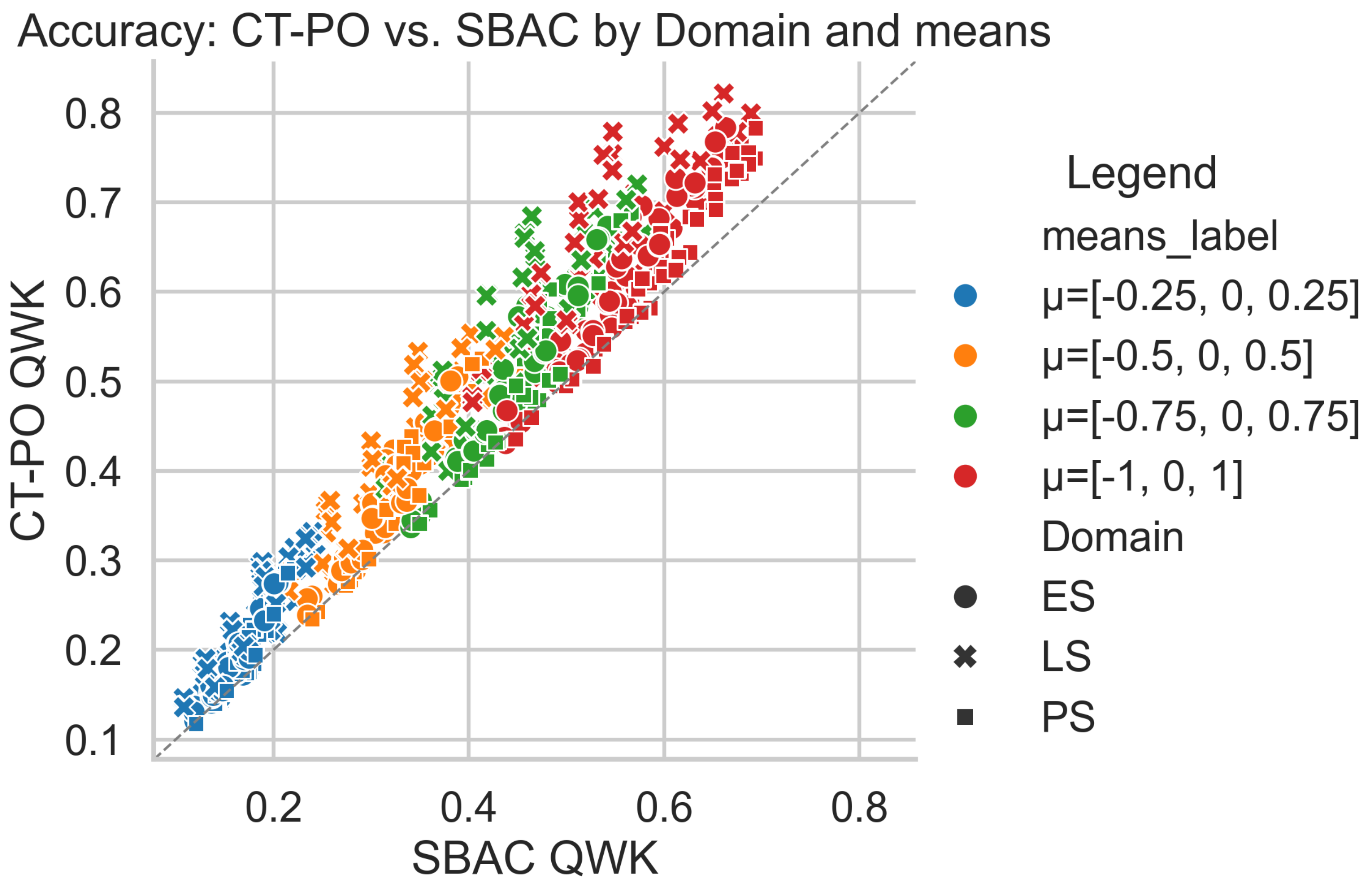
Simulation Conditions: Batch 2

- Mean vectors:
  - [-0.25, 0.00, 0.25]
  - [-0.50, 0.00, 0.50]
  - [-0.75, 0.00, 0.75]
  - [-1.00, 0.00, 1.00]

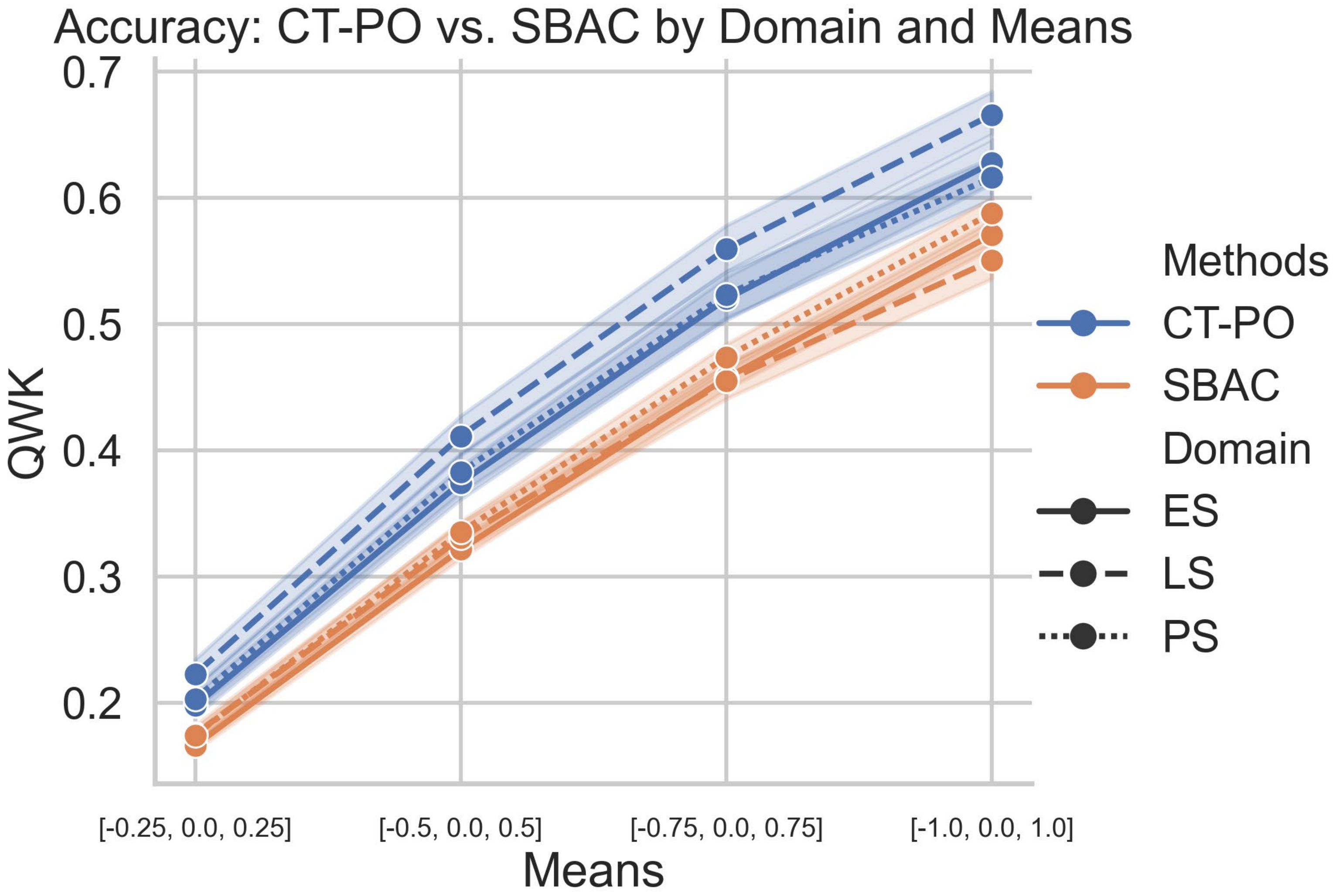
Simulation Conditions: Batch 2

- SD vectors:
  - [0.25, 0.25, 0.25]
  - [0.50, 0.50, 0.50]
  - [0.75, 0.75, 0.75]
  - [1.00, 1.00, 1.00]
  - [0.25, 0.50, 0.25]
  - [0.75, 0.50, 0.75]

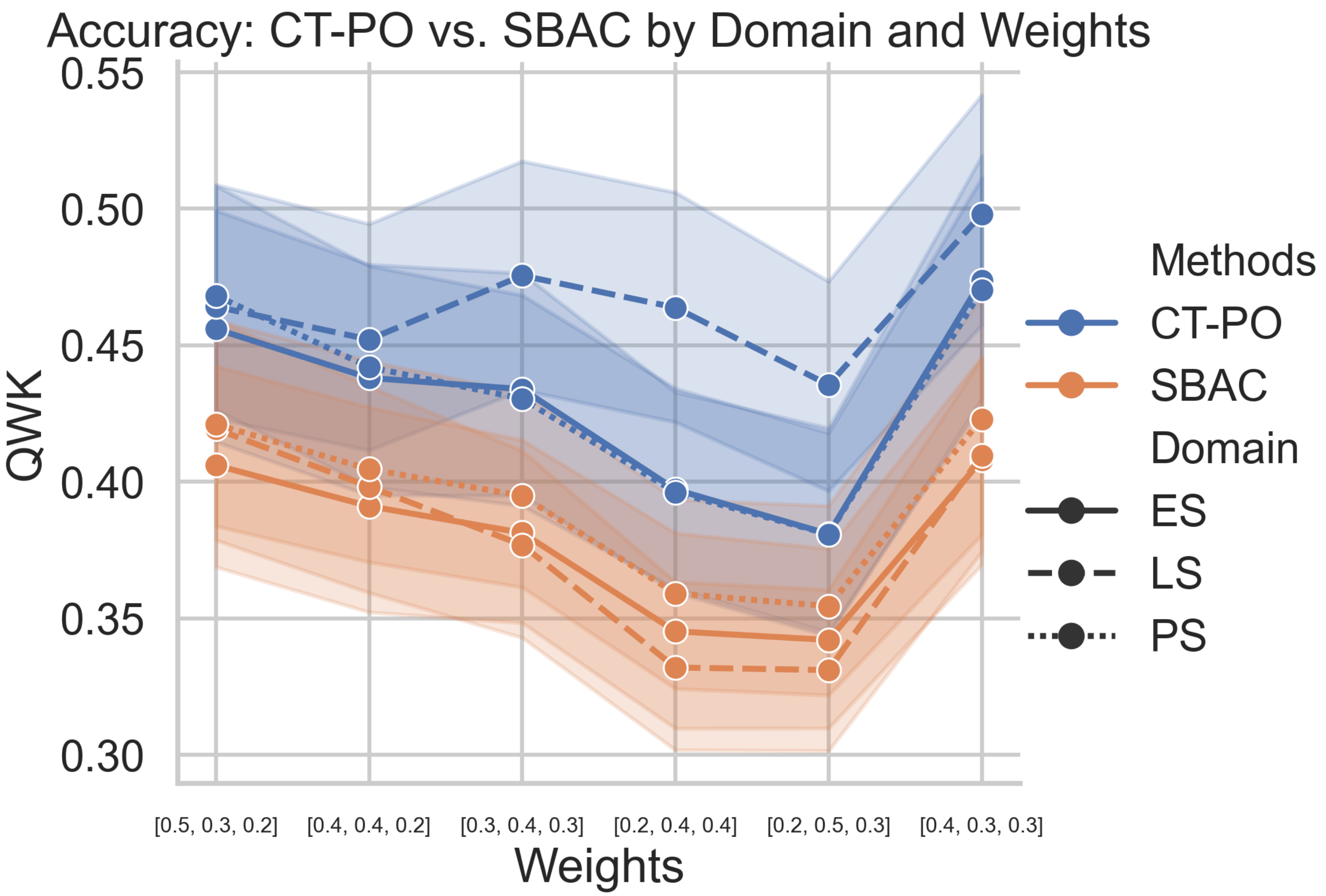
Accuracy: CT-PO vs. SBAC



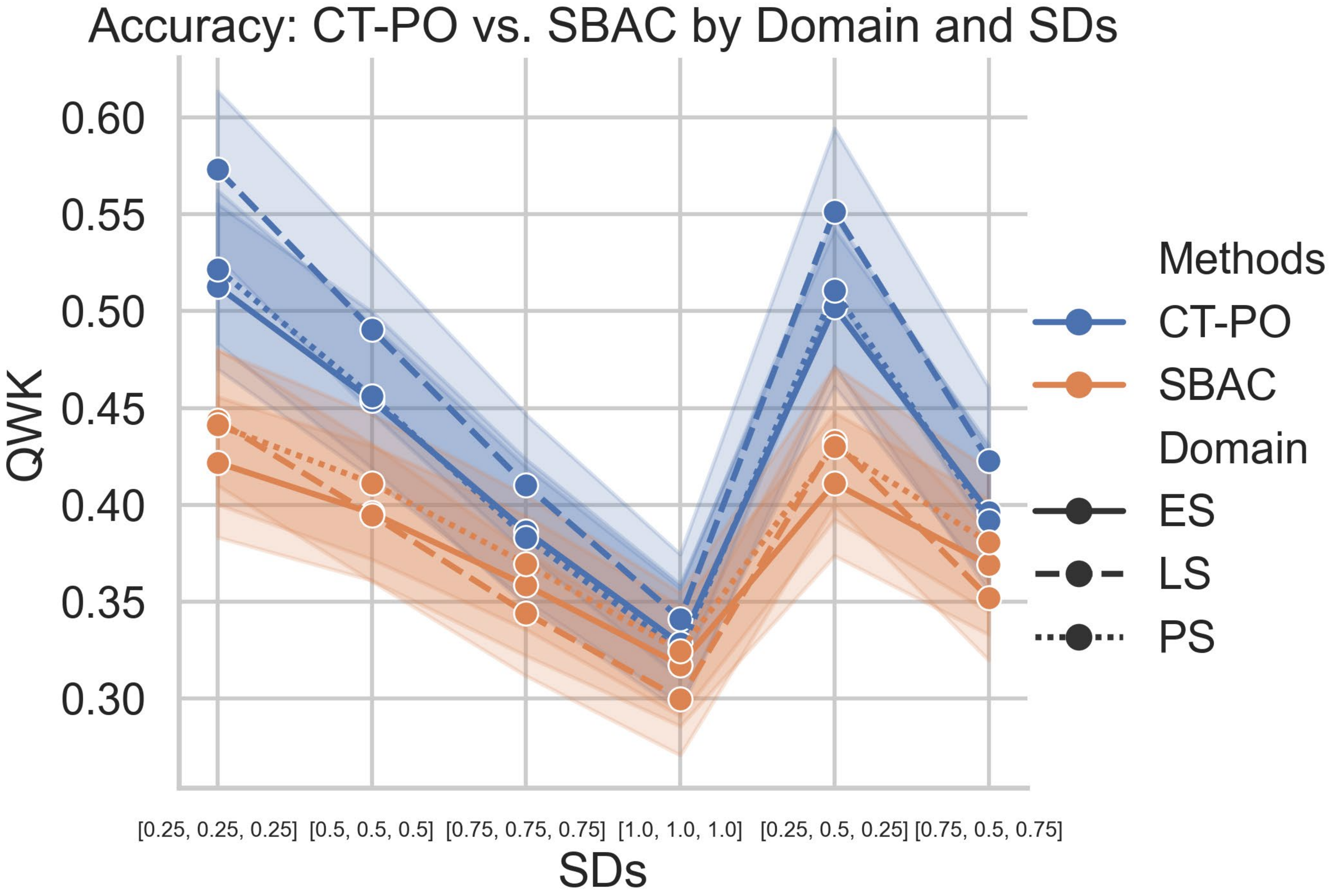
Accuracy: CT-PO vs. SBAC



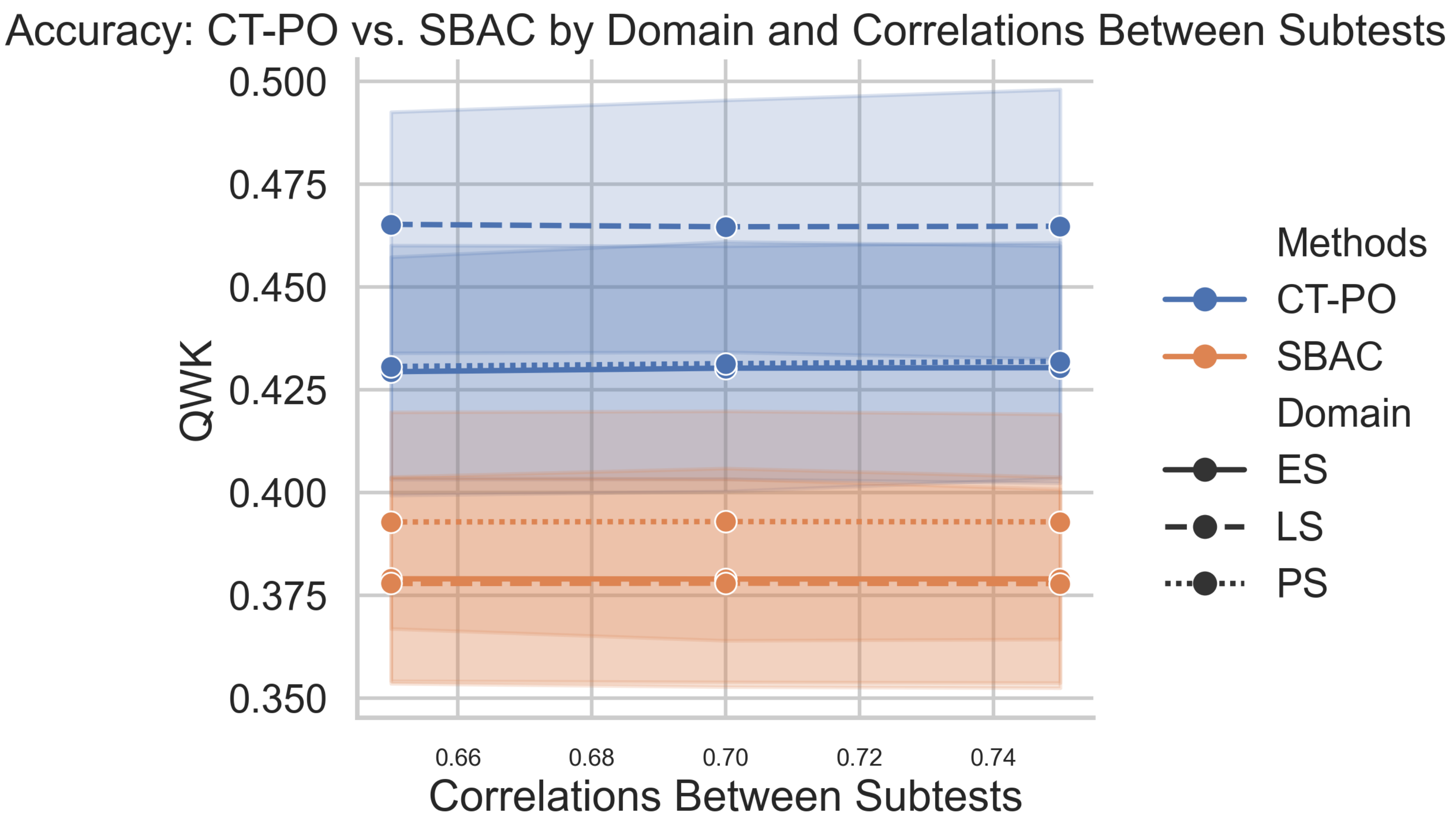
Accuracy: CT-PO vs. SBAC



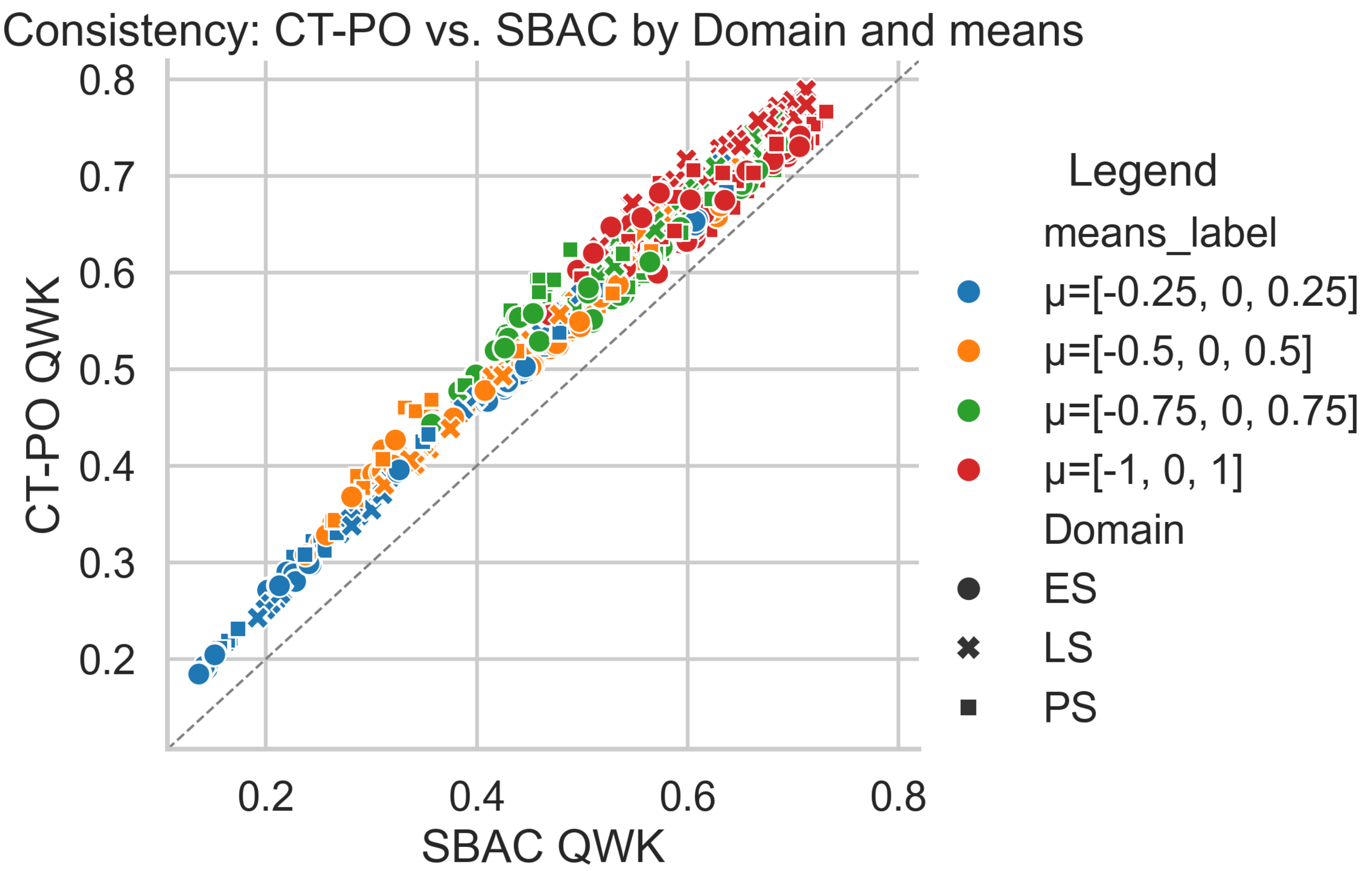
Accuracy: CT-PO vs. SBAC



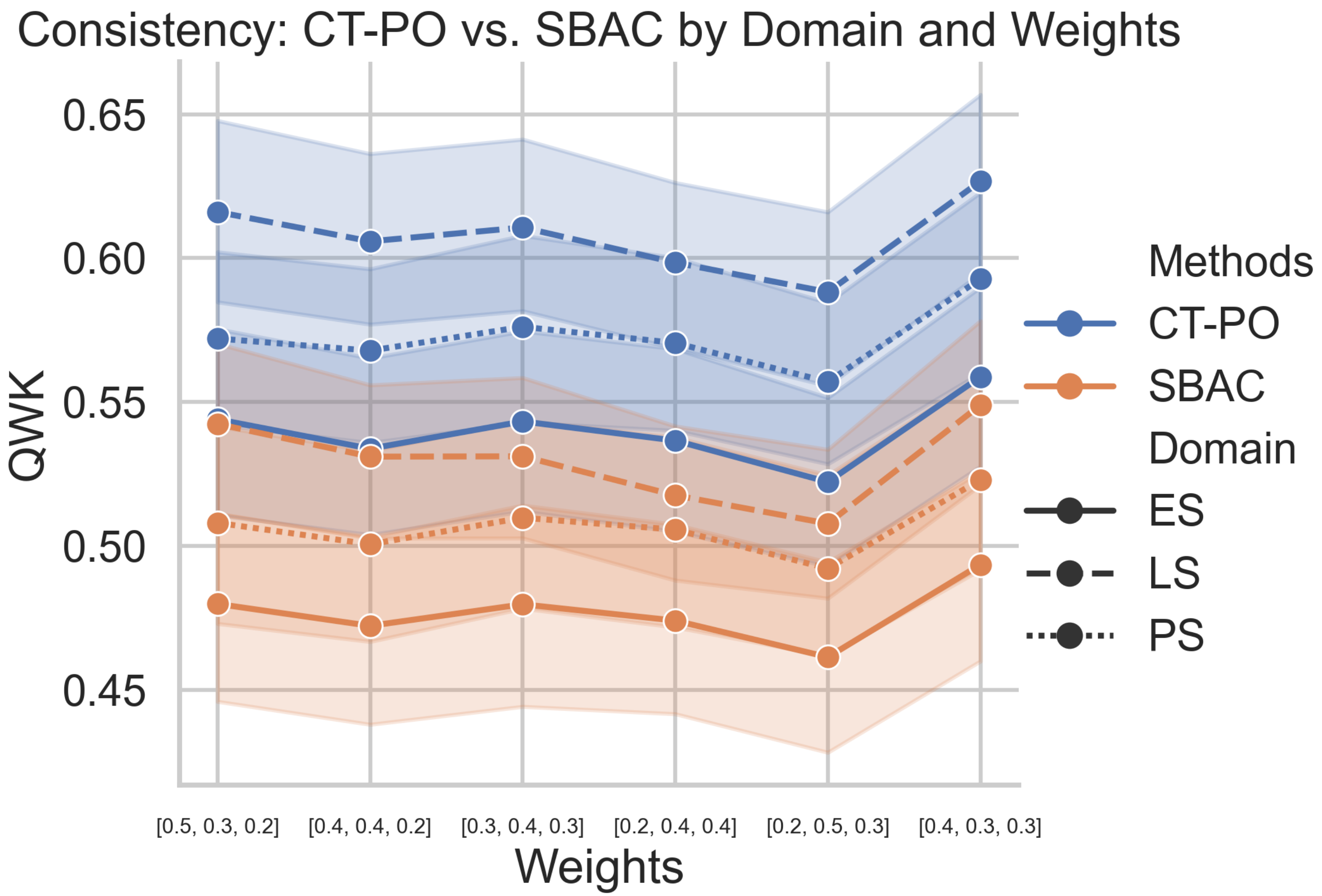
Accuracy: CT-PO vs. SBAC



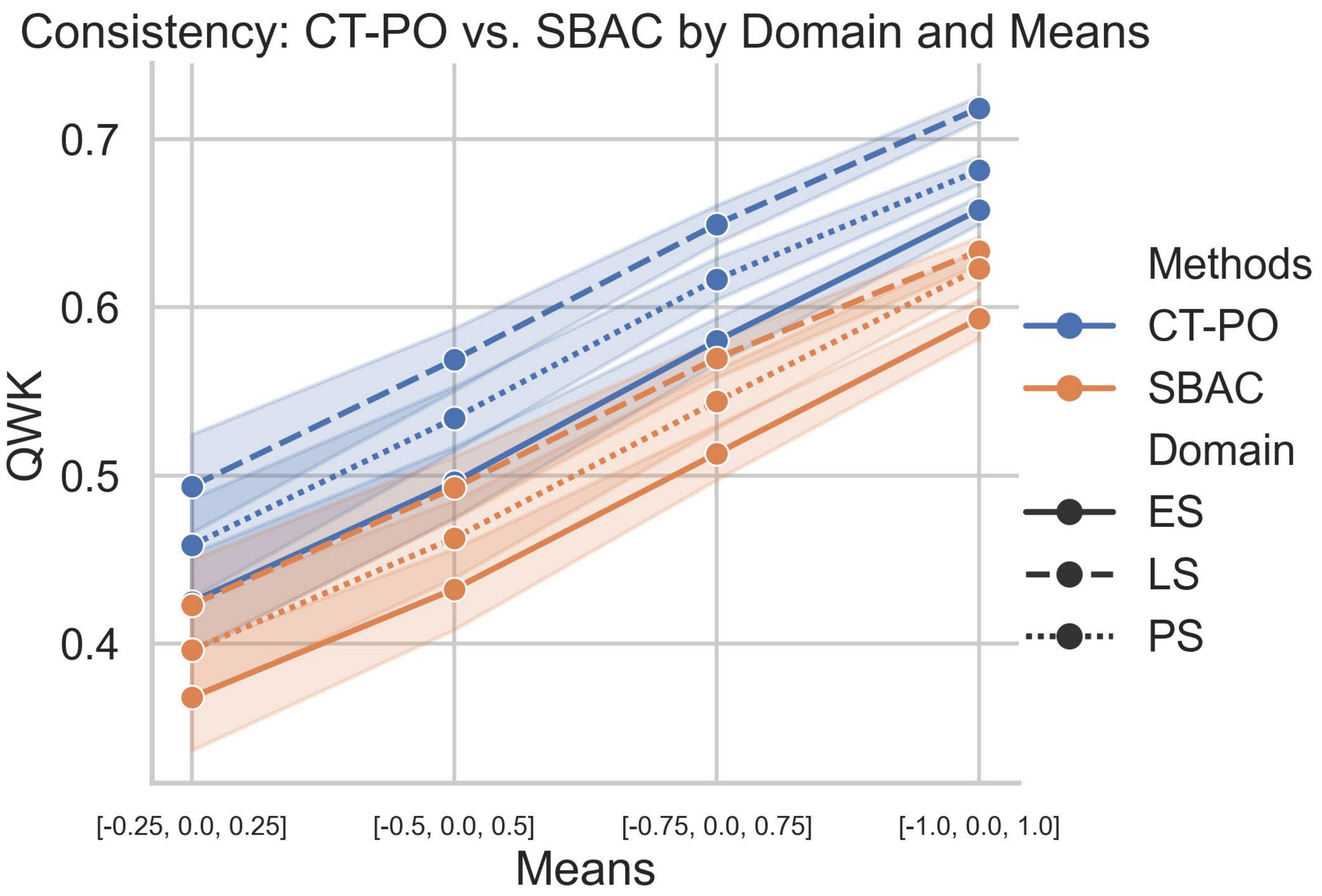
Consistency: CT-PO vs. SBAC



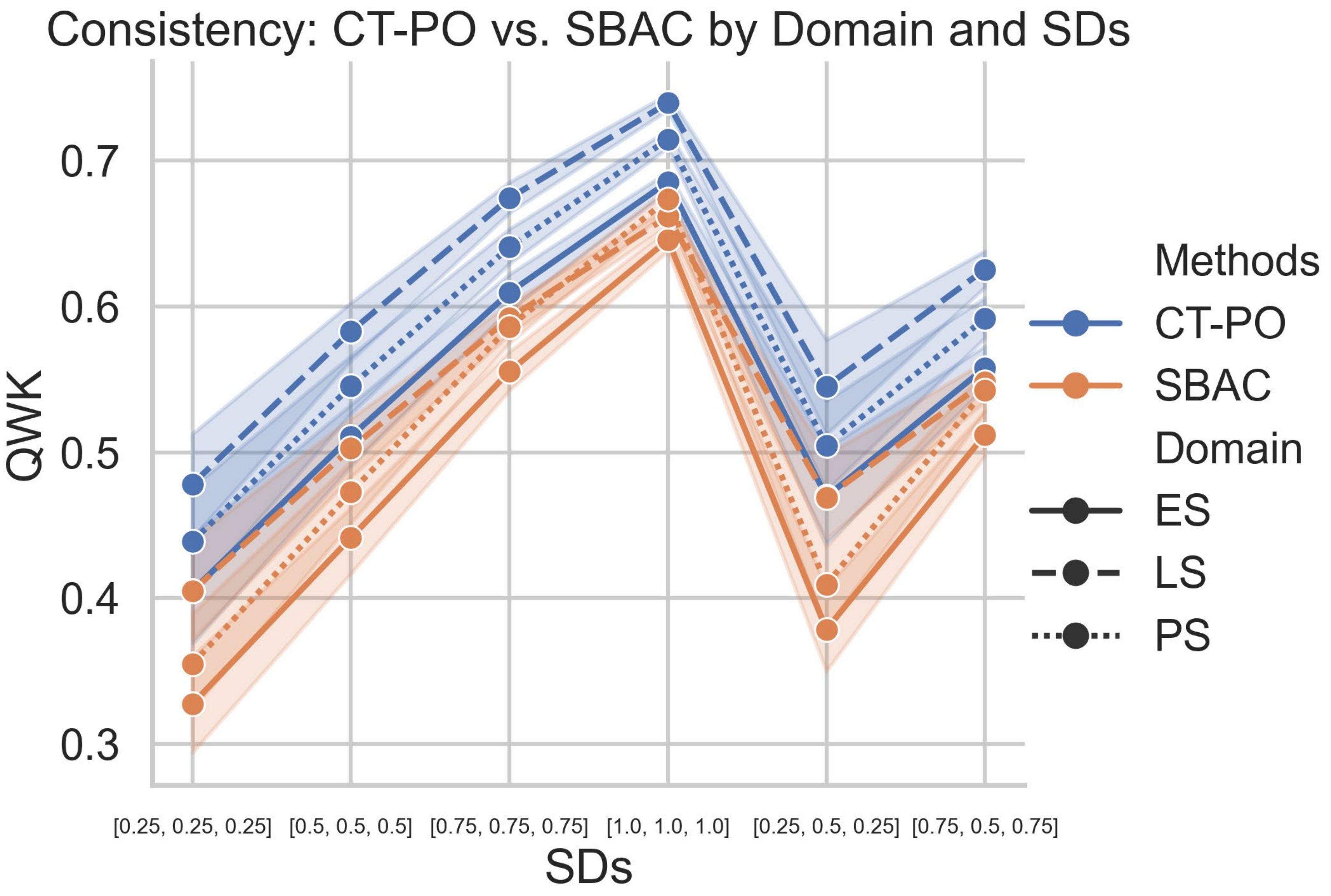
Consistency: CT-PO vs. SBAC



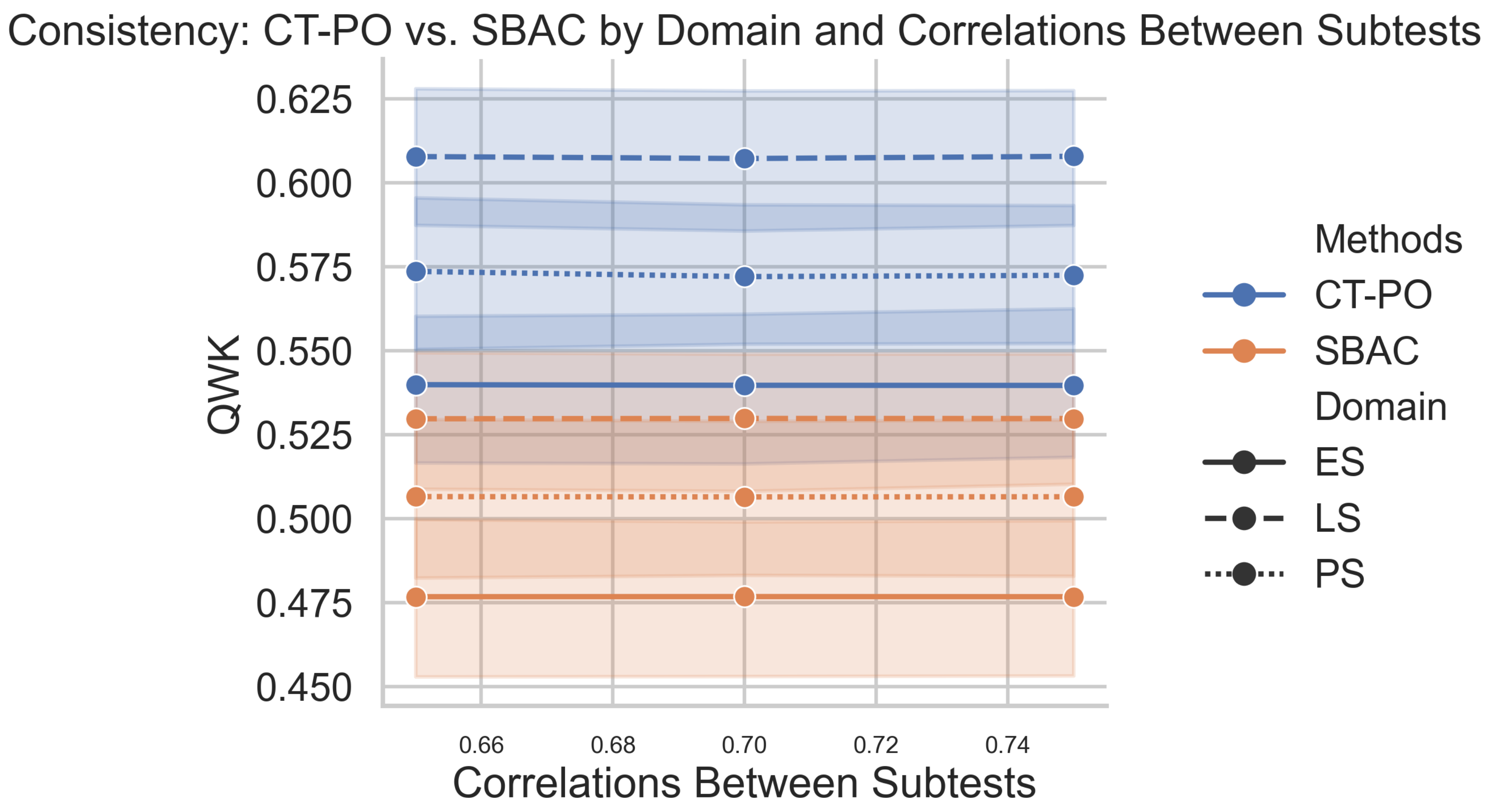
Consistency: CT-PO vs. SBAC



Consistency: CT-PO vs. SBAC



Consistency: CT-PO vs. SBAC



Conclusions and Future Research

- The CT-PO method has shown its advantage over the SBAC method via both the empirical and simulation data
- More simulation conditions will be explored, which include but are not limited to varying item numbers, higher correlations among subtests, different mean and SD vectors, etc.